Calibrated structural analysis of the Indus script: positional classes, slot grammar, a full sign map, and null-model tests of published readings

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Abstract

We re-examine the Indus script using two independent machine-readable transcriptions: Mahadevan's 1977 concordance (M77; 3,573 lines, 13,372 sign tokens, 417 sign types) and an ICIT-derived corpus (Wells/Fuls numbering, distributed in the *lipi* repository). Every quantitative claim is calibrated against a null model: within-line shuffles, global shuffles, permuted keys, length-matched Markov nonsense or memoryless generators, as appropriate. Every claim is also checked on held-out data, on de-duplicated data, or by cross-corpus replication.

1. **Positional classes.** Text-final (terminal), text-initial and pre-terminal sign classes are robust. Their false-discovery rates against shuffles are 0.002, 0.003 and ~0, and 87.5% of terminal-class signs replicate in the second corpus. Thirty-eight signs reach a positional-class confidence of at least 85%; 28 of these also replicate across corpora.

2. **Slot grammar.** A hidden Markov model of seal texts recovers a reproducible slot template: initial → post-initial → medial → terminal → post-terminal. It gains 0.38 nats/token on held-out, de-duplicated texts, against no gain on shuffled texts.

3. **Segmentation.** Byte-pair segmentation lowers held-out description length by about 1 bit per sign after de-duplication, against no reduction for shuffled texts. This shows that recurrent multi-sign units exist.

4. **Linguistic status.** Measures proposed to separate writing from non-linguistic symbol systems do not separate them at Indus text lengths. These are bigram conditional entropy (Rao), Sproat's repetition ratio r/R, and Lee et al.'s Cr/Ur. Linguistic comparanda cut to Indus lengths cross Sproat's boundary, and the two Indus transcriptions disagree with each other (r/R 0.66 vs 0.26). Linguistic status therefore remains **undetermined** by internal statistics.

5. **Published readings.** A sign-value key published by Yajnadevam does not distinguish real from shuffled sign order on any of three measures, including a full Sanskrit morphological parse. Fitted rebus keys built from a Dravidian lexicon perform no better than keys built from English, Finnish or nonsense lexicons.

6. **Phonetic values.** We find **no sign whose phonetic value can be estimated above null**. The stroke numerals are an iconic, plate-checked exception, assessed qualitatively. A complete phonetic decipherment at 85% confidence is not supported by the available evidence.

7. **Full sign map (session 4).** We mapped all 417 M77 signs and 712 ICIT signs. Each entry gives cross-scheme IDs, allographs, positional profile, class, neighbours, sites, object types, glyph references, foreign look-alikes and published claims.

    - A distributional-plus-visual model recovers hand-made M↔W concordance links at 84.7% top-1 under cross-validation (chance 0.25%).

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    - A visual check of 40 Carlson links, made by the project's AI research assistant (not a human epigrapher), judged 37 correct or plausible; an independent human check is still outstanding.

8. **Claims re-tested.** We compiled 42 hand-sourced sign-value proposals and two complete keys, and tested each per sign with held-out values or held-out sites.

    - No phonetic value beats the alternative-value, shuffled-corpus and alternative-lexicon nulls.
    - Yajnadevam's per-sign advantage equals that of a key fitted to a shuffled corpus.
    - Twelve claim rows (nine distinct tests) have their implied **roles** replicate on sites outside Mohenjo-daro and Harappa: jar and arrow are final, M267 and M150 are initial, and numeral+fish, M087+fish and M267+M099 are units. These results do not discriminate between proposed sound values.

9. **Look-alikes.** Visual look-alikes in Linear Elamite, Egyptian, cuneiform, Brahmi and Linear A do not predict Indus sign function beyond the trivial resemblance of stroke numerals.

10. **Prior work re-run (session 5).** We replicated two published results on our data and added baselines they lacked.

- Nair (2026): Zipf slope −1.40 to −1.44 (published −1.49) and bigram conditional entropy 3.43–3.60 bits (published 3.23), both far below within-inscription permutations ($p = 0.001$).
- Yadav et al. (2010): the big unigram→bigram drop in held-out perplexity replicates (74.8 → 31.3; published 68.8 → 26.7). The further trigram and quadrigram gains do not appear on deduplicated texts in our implementation. Their 74% restoration "sensitivity" is matched by a frequency-only baseline (0.88), because the 90%-mass candidate set holds about 120 signs. Top-1 restoration is the informative figure: 0.34 for a bigram model against 0.12 for frequency alone.
- A synthetic, non-linguistic emblem system (after Raghavendra 2026) passes every structural test we use: low conditional entropy, a bigram perplexity drop, restoration well above baseline and a held-out HMM slot gain. These tests therefore show **ordered structure, not language**, which is how we report them.

1. Introduction

The Indus script (c. 2600–1900 BCE) survives on several thousand short inscriptions, chiefly seals, sealings and miniature tablets. Average lines are about four signs long (Mahadevan 1977; Rao 2010). There is no bilingual. Parpola (1994: 48) puts the methodological consequence directly: "in the absence of effective aids, such as extensive bilinguals, native lexica or descendants of the signs with known values, complete decipherments of logo-syllabic writing systems are usually impossible."

Statistical work has split into two camps:

- **Arguing for linguistic structure:** Yadav et al. 2009; Rao 2010.
- **Arguing that the inscriptions are a non-linguistic symbol system:** Farmer, Sproat & Witzel 2004; Sproat 2014.

Meanwhile, decipherment claims continue to appear. They include Dravidian (Parpola 1994; Mahadevan 2015; Mukhopadhyay 2019, 2021) and Sanskrit (the Yajnadevam *lipi* repository).

Our aim is narrower than decipherment. We ask what can be said about the script at a **stated, calibrated confidence**, and we test published readings with the same machinery. We deliberately separate three things:

- **(a) distributional facts.** These can be given probabilities against explicit nulls.
- **(b) functional interpretations**, e.g. "numeral" or "terminal marker". These are partly calibratable.
- **(c) phonetic or semantic values.** These are calibratable only if they beat null models, and in our tests none do.

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1.1 Related work and what is new here

Table R1 compares this study with published work making similar claims. The full matrix is in `outputs/compare/prior_work_matrix.csv` (24 rows). Cells come only from sources we opened; where we saw only an abstract, the cell says so.

Work	Corpus	Nulls	Held-out	Multiple-testing	Cross-corpus	Code	Main claim
Koskenniemi 1981	n/s	n/s	n/s	n/s	n/s	n/s	syntactic segmentation framework (abstract only)
Parpola 1994	CISI/M77-era	none	none	none	none	n/a	segmentation, classes, Dravidian readings
Farmer, Sproat & Witzel 2004	whole corpus	comparative	none	none	none	n/a	non-linguistic system
Rao et al. 2009 (PNAS)	M77 subset	n/s	perplexity	n/s	none	n/s	Markov structure; restoration
Rao et al. 2009 (Science)	M77 subset	DNA, protein, Fortran, artificial	none	none	none	n/s	entropy “like language”
Yadav et al. 2010	EBUDS 1,548	random equiprobable	5-fold CV	n/s	none	n/s	quadrigram saturation; restoration 74%
Fuls 2013	ICIT 3,723 texts	n/s	n/s	n/s	n/s	DB gated	7 positional classes (abstract only)
Sproat 2014	many real non-linguistic corpora	real systems	n/s	n/s	yes	n/s	entropy measures non-discriminating
Mukhopadhyay 2019	M77/ICIT observations	none	none	none	n/s	n/a	data-carrier syntax, not phonetic spelling
Nair 2026	1,916 dedup texts	2 generators + 7 real corpora; 1,000 perms	sweeps	Bonferroni (variant)	no	public	Indus intermediate
Raghavendra 2026	SIGIL 3,000 + Indus	synthetic emblem system, 54 methods	grouped held-out	Bonferroni	n/s	corpus released	measures non-specific
This work	M77 3,573 lines + ICIT 712 signs	shuffles, fitted/noise keys, alt lexicons, synthetic emblem	dedup CV, LOO, site-held-out	BH-FDR	yes (scored concordance)	scripts	structure yes; language undetermined; no value beats null

n/s = not seen in the parts we read. Computational decipherments of other scripts (Snyder, Barzilay & Knight 2010; Luo, Cao & Barzilay 2019; Luo et al. 2021) are validated against known answers (Ugaritic, Linear B, Gothic). They need a known related language and, except Luo et al. 2021, word boundaries; Indus has neither. Image-recognition work (Palaniappan & Adhikari 2017; Dixit et al. 2025; Ganeriwala et al. 2025) solves a different task. A visual-similarity study (Reddy 2025) reports cosine similarities without a null; our look-alike tests (§4.11) suggest such resemblances do not carry function. A press survey (Roychowdhury, *Indian Express*, 12 Aug 2026) reports about 200 claims emailed to Rao and at least 60 to RMRL, and no agreed mechanism for evaluating them.

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Where prior work is stronger.

- Fuls and Wells work from the full curated ICIT database, with a finer 695-sign list and archaeological context; we use a secondary, unlicensed derivative.
- Sproat (2014) and Nair (2026) use real non-linguistic corpora; we do not have them.
- Raghavendra (2026) registers 54 methods against a synthetic control; our synthetic control covers only our own tests.
- Image-based recognition is outside our scope.

What is new here. As far as we can tell from the works we opened, no earlier study combines these five things:

1. Class membership reported as a calibrated probability (bootstrap stability × (1 − FDR against shuffles)), replicated across two independent transcriptions linked by a scored concordance (84.7% top-1 under cross-validation).
2. Duplicate-aware held-out tests for segmentation and slot grammar. Yadav et al. (2010) already removed duplicates; we add frequency and position baselines and report top-k rather than mass-set sensitivity.
3. Per-sign leave-one-out tests of published phonetic values against alternative-value, shuffled-corpus, alternative-lexicon and noise-fitted keys.
4. Site-held-out role tests: roles are learned on Mohenjo-daro and Harappa and tested on 641 texts from other sites, with BH correction.
5. A full M/W sign map with null-calibrated foreign look-alike tests.

None of these settles linguistic status, which our results, Sproat (2014) and Raghavendra (2026) agree internal statistics cannot do.

2. Data and provenance

Corpus	Source	Units used	Access / licence
M77	Mahadevan 1977 concordance, machine-readable export (joyboseroy/indus_decipher)	3,573 lines; damage marker splits lines; 2,906 texts	ASI copyright; research use only
M77 object codes	Our OCR of the IA scan of Mahadevan 1977 (6-digit type/side/field/line/direction codes)	2,601 of 2,906 texts (89.5%)	derived; research use only
ICIT-derived	*lipi* repository `inscriptions.csv` (commit fc8813a), Wells/Fuls numbering	5,680 rows; 18,050 tokens; 712 sign codes	no licence; ICIT data are not open; private research use
Concordance M ↔ P ↔ W	M. Carlson's hand matches (mayig)	1,532 rows	unverified, many-to-many
Weights	Sproat, `weights.xml` (Hemmy 1931–43; HARP)	1,154 weights	GPL-3
Dravidian lexicon	Merriam & Fuls, *Dravidian Database* v1.2 (Zenodo 22682542)	DEDR forms + 15,629 PDR	CC BY 4.0
Sanskrit lexicon	Monier-Williams via csl-orig; vidyut 0.4 data	172k reduced headwords; parser	CC BY-SA 4.0; MIT
Other lexicons	FrequencyWords (fi, tr, id, hu, en)	50k lists	CC BY-SA 4.0 content
Comparanda texts	Random Wikipedia articles (ta, sa, fi, en), ~60 per language	76k–277k characters	CC BY-SA 4.0

Transcription rules.

- M77 lines are in reading order. ICIT lines are stored in visual order and were reversed for reading.
- Damage codes (M77 `MSg0`; ICIT 000/999) split a line into undamaged stretches. "Complete lines" means undamaged lines with no split.

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- Exact duplicate texts are common: 35% of M77 complete lines and 39% of ICIT complete lines. This inflates held-out scores (Bose 2026), so the key analyses were repeated on de-duplicated corpora.

OCR of M77 object codes.

- Glyphs were classified by a k-nearest-neighbour digit model trained on 17,000 automatically labelled glyphs.
- Held-out-page digit accuracy was 0.997–0.999, and agreement with an independent OCR pass was 99.3% on 1,822 overlapping texts.
- Single-sign direction codes matched line length in 53 of 53 checkable cases, with no false positives in 1,785.

Plate checks. The stroke numerals and fish signs were checked visually against the M77 sign-list plates (book pp. 32–38) by the project's AI research assistant, not by a human epigrapher. Crops are kept privately in `paper/evidence/` and are not for publication.

3. Methods

All code is in `scripts/` and all outputs in `outputs/`; file names are given in brackets.

3.1 Positional classes (`structure.py`)

For each sign with frequency ≥ 10 in M77 we computed shares of initial, final and solo occurrences, plus left/right context entropies. Classes were assigned by a fixed decision list, using one-sided binomial tests ($\alpha = 0.001$) against corpus baseline shares:

Class	Rule
TERMINAL	final share ≥ 0.5 and significant
INITIAL	initial share ≥ 0.5 and significant
PRE-TERMINAL	≥ 50% of successors are terminal-class signs
STAND-ALONE	solo share ≥ 0.15 and ≥ 5 solo uses
BOUND-COLLOCATE	top right collocate ≥ 60%
MEDIAL/UNSPECIFIED	otherwise

Calibration:

- **False-discovery rate per class:** the mean class count over 30 within-line shuffles, divided by the real count.
- **Stability:** the share of 100 bootstrap resamples of lines that give the same class.
- **Reported distributional confidence** = stability × (1 − FDR).
- **Replication:** the same rules applied to the ICIT corpus through the concordance, with positional classes compared.

Stroke signs were also tagged NUMERAL-STROKE by visual inspection (made by the AI research assistant, not a human). This tag is kept separate from their positional class.

3.2 Segmentation (`structure2.py`, `structure3_dedup.py`)

1. Byte-pair merges (K = 0–800) were learned on 80% of lines.
2. Held-out cost was scored as bits per sign under a unigram unit model, with unseen units spelled out sign by sign (add-½ smoothing).
3. The same procedure was run on within-line-shuffled corpora, and on de-duplicated corpora. Results are averaged over 3 splits.

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3.3 Slot grammar (`structure2.py`, `structure3_dedup.py`, `structure_ling.py`)

- **Model:** categorical HMMs (hmmlearn), K = 1–6 states, fitted to complete seal texts (object type 1). Signs with training frequency < 5 were mapped to UNK, and an END symbol was appended to each text.
- **Main test:** held-out log-likelihood per token, real vs within-text shuffled.
- **Slot-order conformity test:** for K = 6 on de-duplicated seals, states were ranked by their mean relative position in training texts. For each held-out text we checked whether the Viterbi path is non-decreasing in slot rank. This was compared with held-out texts that were shuffled or reversed.

3.4 Script-type and linguistic-status measures (`structure_ling.py`)

Inventory and coverage. We report observed types, hapax counts, the Chao1 unseen-type estimate, and the number of types covering 80%/90% of tokens.

Status measures. Each was computed on 20 random subsamples of 9,000 tokens per system. Ranges are 2.5–97.5 percentiles over the subsamples, which reflect subsampling variability rather than a full bootstrap. The measures are:

- **Conditional entropy:** bigram H(next | prev) with boundary symbols and Miller–Madow correction. It is reported as a ratio to the same system after a global token shuffle with text lengths preserved. This ratio is 1 for a memoryless system and near 0 for a rigid one.
- **Normalised block entropies** B2 and B3 (cf. Rao 2010). We used the Miller–Madow estimator, not the NSB estimator of Nemenman et al. (2001) used by Rao and by Sproat, so B3 is undersampled.
- **Sproat's (2014) repetition ratio r/R.** R counts tokens repeating an earlier token in the same text; r counts those repeats that are adjacent to it. Sproat's linguistic/non-linguistic boundary is 0.10.
- **Lee et al.'s Ur and Cr** (a = 7), with their decision tree as quoted by Sproat (2014): Cr ≥ 4.89 counts as linguistic; Ur < 1.09 as segmental, < 1.37 as syllabic, otherwise logographic.

Comparanda.

- Linguistic texts: Tamil and Sanskrit graphemes (Unicode extended grapheme clusters ≈ aksharas), Finnish and English letters, and words in all four languages. Each was cut into windows whose lengths are drawn from the M77 line-length distribution ("length-matched"). Natural-sentence versions of Tamil aksharas and English words are included for reference.
- Non-linguistic baselines, all built from Indus unigram frequencies: memoryless Zipf; "merit-badge" texts without repetition (Sproat's example); and a rigid deterministic-successor model.

3.5 Claim harness (`claim_harness.py`, `parse_test_vidyut.py`, `rebus_harness.py`)

Yajnadevam (lipi). The key (716 sign entries, 129 distinct values) and the reading rules were taken from the repository's code. Four tests:

- **A. Reproduction of the published readings.**
- **B. Fixed-key segmentability** into Monier-Williams headwords and length-stratified lexicons.
- **C. Fitted keys** (hill-climbing over the same value inventory) for 8 lexicons, on held-out texts.
- **E. Full Sanskrit parse with vidyut**, comparing real order with shuffled order, a permuted key and Markov nonsense.

Rebus/Dravidian.

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- Keys mapping the top 150 signs to 2–4-character lexical roots were fitted to maximise the share of adjacent sign pairs whose concatenation is an attested word. Scoring was on held-out texts, for each of 8 lexicons (2 seeds × 3 corpora: real, globally shuffled, within-text shuffled).
- Parpola's specific numeral + fish compounds were looked up in the Dravidian DB.

3.6 Anchors (`anchors.py`, `w700_weights.py`)

- Numeral value frequencies.
- The tablet formula: long-stroke numeral + W700.
- The weight ratio classes (Hemmy unit).
- Motif and site enrichment, with max-statistic permutation correction.
- West-Asian/Gulf sequences vs the Indus bigram space.
- Duplication by object type.

3.7 Full sign map and concordance scoring (`concordance.py`, `concordance_relabel.py`, `sign_map_full.py`)

Glyph images.

- Every M77 sign was cropped from the Mahadevan (1977) sign list (pp. 32–35): 416 of 417 signs; M049 is missing. Crops are placed from the OCR position of each row's labels.
- Every Wells-numbered sign was rendered from the SK Indus Script font (631 glyphs).
- Both image sets are **private research use only** (ASI copyright; the font is "All Rights Reserved").

Visual similarity. Glyphs are binarised and normalised to 48 × 48 pixels. Similarity is 0.5·exp(−chamfer/3) + 0.5·HOG cosine, taking the maximum over a horizontal mirror.

Distributional profile. Each sign is described by:

- solo, initial and final shares
- log frequency
- left- and right-neighbour vectors, projected onto anchor signs

Anchors and cross-validation.

- Anchors are the 118 one-to-one Carlson links with frequency ≥ 3 in both corpora.
- A logistic model on five features (left/right cosine, position difference, frequency difference, visual similarity) was trained with 5-fold cross-validation.
- It was evaluated by top-1 and top-5 accuracy among about 399 W candidates; chance top-1 is 0.25%.
- Per-fold and global Hungarian assignments were also computed.

Scoring all 566 Carlson links.

- Every link was ranked among the W candidates, with the other hand-matched variants of the same M sign excluded.
- Rare signs (no distributional rank) received a visual-only rank among all 631 W renders.

Visual check (AI-made). Forty links were inspected visually, M crop beside W render: the 20 the model scores lowest plus 20 at random. The inspection was made by the project's AI research assistant (an image-capable language model), **not by a human**. It is therefore not independent expert validation; a blind human check is planned (§6).

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3.8 Claims table and per-sign leave-one-out tests (`claims_table.py`, `loo_signs.py`, `loo_hand.py`)

Claims table. We compiled every sign-value proposal we could source from documents opened in this project:

- Parpola (1994: 59–60, 181, 187–188, 194, 197, 215, 264, 276–277). These pages include his reports of the Knorozov team, Mahadevan's early work, Fairservis and Heras.
- Mahadevan (2015)
- Wells (2015, a caption only)
- Mukhopadhyay (2019, class-level only)
- the complete Yajnadevam key (714 signs, Sanskrit)
- an unsupervised optimal-transport key by Field Cady (99 signs, Dravidian syllables)

Sign mapping. Signs named only in figures were mapped to M numbers by glyph similarity on crops of the source chart. Each mapping carries a confidence.

Phonetic keys. For each sign in ≥ 5 unique clean ICIT texts, the proponent's value was held out. The statistic is the share of that sign's occurrences lying inside an attested word that spans a neighbouring sign. Attested words are those of ≥ 4 characters in a length-matched lexicon.

- The proponent's value was compared with every other distinct value in the same key (exact rank test). The rest of the key was held fixed.
- The statistic was then recomputed on within-text shuffles and with 7 alternative lexicons.
- It was also computed for two calibration keys produced by the same hill-climb: one fitted to a shuffled corpus, and one fitted to the real corpus from a random start.

Logographic and morphemic claims. These cannot be scored lexically. Instead we tested the distributional prediction each claim implies:

- place sign → text-initial
- suffix → text-final
- compound → adjacent pair
- paradigm alternants → shared left contexts

The tests used 641 ICIT texts from sites **other than Mohenjo-daro and Harappa**, the material on which the claims were mostly built. The null was 2,000 within-text shuffles. These tests check the role a claim implies, not its sound value.

3.9 Branching entropy and NSB entropy (`structure4_be.py`, `nsb_entropy.py`)

Branching-entropy segmentation.

- Boundaries were placed where forward + backward branching entropy increases (the branching-entropy criterion associated with Tanaka-Ishii; that paper was not opened in this project), using contexts of up to two signs.
- The model was trained on 80% of the 1,251 de-duplicated M77 seal texts.
- Its held-out boundaries were compared with BPE boundaries (K = 400) and with HMM slot boundaries (K = 5). The null was random boundaries at the same per-text rate.

NSB entropy. NSB estimates (Nemenman, Shafee & Bialek 2001) were computed by our own implementation, because the `ndd` package failed to build. We report:

- unigram entropy at three alphabet sizes

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- bigram conditional entropy, against within-text shuffles

3.10 Comparative look-alikes (`render_comparanda.py`, `lookalikes.py`)

Repertoires compared.

Script	Glyphs	Source	Licence
Linear Elamite	394	elamicon FontForge sources	GPL-2
Egyptian hieroglyphs	1,076	Noto	OFL
Cuneiform	1,229	Noto (Unicode, Neo-Assyrian-style, not archaic forms)	OFL
Brahmi	92	Noto	OFL
Linear A	341	Noto (added comparandum)	OFL

We found no open glyph source for Proto-Elamite, archaic proto-cuneiform or South Indian megalithic graffiti, so these are **not covered**.

Tests. All matches are **visual resemblance only**; no reading is implied.

1. Mutual information between the category of each M sign's best look-alike and its Indus structural class, against 10,000 label permutations. Categories are Gardiner group (Egyptian), vowel/consonant/numeral (Brahmi), AB/A-only/fraction (Linear A) and numeric/sign (cuneiform). The test was run with and without stroke numerals.
2. An Indus-internal Mantel test of visual similarity against positional-profile similarity.

3.11 Replications of prior work and a synthetic control (`compare_replications.py`, `compare_sigil_hmm.py`, `prior_work_matrix.py`)

- **Yadav et al. 2010.** An EBUDS-like corpus: undamaged single-line M77 texts, one copy per duplicate (1,849 texts against their 1,548; their damage filter was stricter).
  - Interpolated Witten–Bell n-grams (n = 1–5), 5-fold CV, held-out perplexity including the end token.
  - Restoration: one random sign is dropped per held-out text (length ≥ 2), and every candidate sign is scored by full-text probability.
  - Reported: Yadav's "sensitivity" (truth inside the smallest set holding 90% of the mass), the set size, and top-1 and top-5.
  - Added baselines: frequency-only (unigram) and position-aware unigram scorers.
- **Nair 2026.** OLS fit of log frequency on log rank over all ranks. Plug-in bigram conditional entropy within inscriptions, against 1,000 within-inscription permutations (as in Nair). Run on deduplicated ICIT/lipi texts and on M77.
- **Fuls 2013 (approximation).** Fuls' exact formula was not available to us; we saw only the abstract. Each sign's mean relative position $i/(L-1)$ over texts of length ≥ 2 (signs with ≥ 10 tokens) is binned into initial (≤ 0.15), early-, mid- and late-medial, and terminal (≥ 0.85), then cross-tabulated with our classes.
- **SIGIL-like control (after Raghavendra 2026).** Six ordered fields (group, title, commodity, quantity, office, tag) with Zipfian vocabularies (362 signs in total).
  - Each field's value depends on the previous field's value through sparse Dirichlet tables; optional fields are dropped to match M77 text lengths.
  - The system has no phonology and no words.
  - The same perplexity, restoration, conditional-entropy and HMM tests are applied.

4. Results

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4.1 Positional classes are real and replicate

Of 153 signs with frequency ≥ 10, the classes break down as: 68 medial/unspecified, 20 pre-terminal, 20 initial, 18 numeral-stroke, 16 terminal, 9 bound-collocate and 2 stand-alone (Fig. 1). Shuffled corpora produce almost none of the non-medial classes:

Class	Null count (mean of 30 shuffles)	FDR	ICIT replication of positional class
TERMINAL	0.03	0.002	14/16 (87.5%)
INITIAL	0.07	0.003	52%
PRE-TERMINAL	0.0	0.0	64%
BOUND-COLLOCATE	0.17	0.019	33%
STAND-ALONE	4.0	1.0 (not real)	0%

Key signs:

- **Terminal:** the jar M342/W740 (1,395 tokens, 70.2% final, $p \approx 10^{-272}$), M176 (89% final), M328/W700 (85%), M211 (82%) and M012 (87%).
- **Initial:** M267 (79.7% initial; Mahadevan's "fortified house"), M391, M204 and M150.

The pre-terminal class is cohesive in context space (p = 0.0008). That result is partly built into its definition, which is based on what follows each sign.

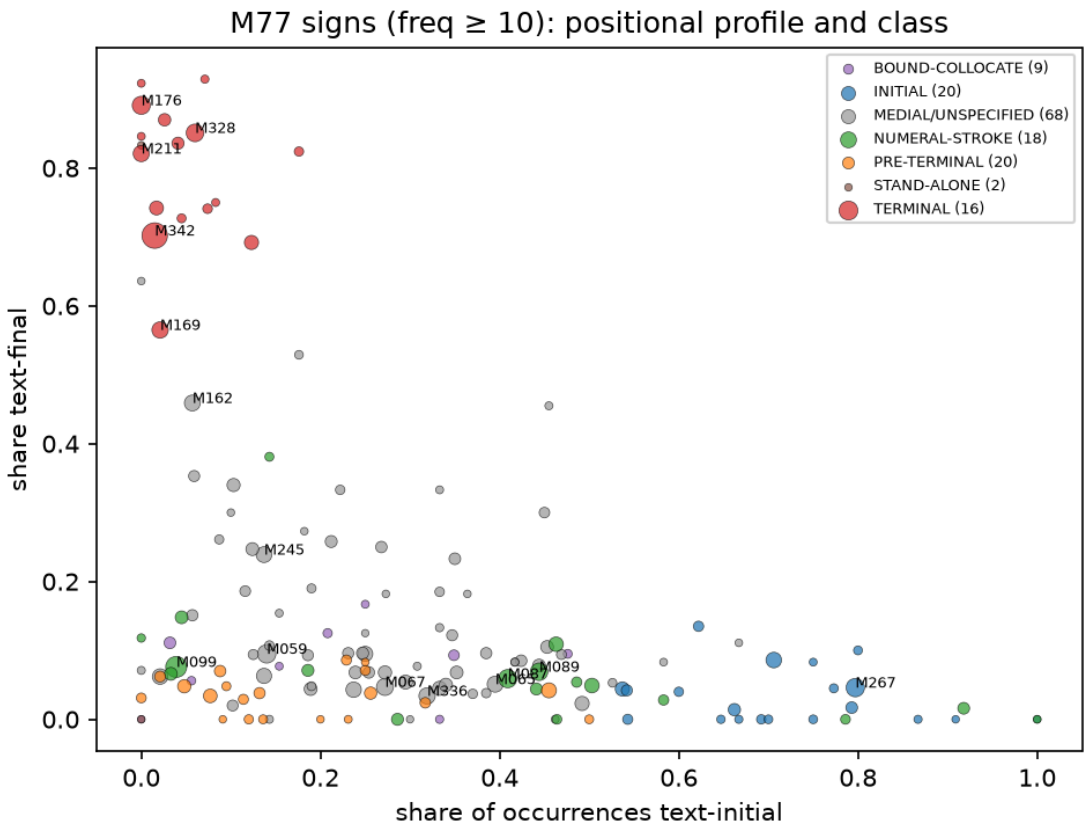


Fig. 1. Positional profile of M77 signs (freq ≥ 10), coloured by class; marker area ∝ √frequency.

The stroke numerals are **not** a distributional class. Their context vectors are no more alike than random frequency-matched sets (short strokes p = 0.70; long strokes p = 0.74). Their numeral status therefore rests on iconicity, which we checked on the plates, and on substitution within formulae (§4.5), not on position.

4.2 Segmentation: recurrent multi-sign units

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Held-out description length falls with BPE merges for real texts but not for shuffled ones (Figs. 5, S1):

Corpus	K = 0	K = 400 (real)	K = 400 (shuffled)
M77 complete lines	7.32	5.72	7.21
M77 de-duplicated	7.60	6.60	7.65
ICIT de-duplicated	7.93	7.16	8.02

The gain after de-duplication (≈1.0 bit/sign in M77, 0.8 in ICIT) shows that the units are not just repeated whole texts. The most frequent learned units are:

- M267+M099 (123)
- M089+M328 (120), M087+M328 (77) and M095+M328 (55): long-stroke numeral + W700 tablet formula
- M391+M099 (57)
- M048+M342+M176 (46)
- M347+M342 (40)

We did not formally compare segmentation against branching-entropy boundaries. The units are statistical chunks, and they need not correspond to words.

4.3 A slot grammar for seal texts

On de-duplicated seal texts (n = 1,251), held-out log-likelihood per token improves with the number of HMM states for real texts but not for shuffled texts:

Text set	Gain vs K = 1 at K = 2–5 (nats/token)
Real	+0.10, +0.11, +0.31, +0.38
Shuffled	+0.01, −0.03, −0.01, −0.04

Across three random splits, the K = 6 model consistently recovers this order:

1. **INITIAL:** M267, M391, M293, M150
2. **POST-INITIAL:** M099, M123, M343, M097
3. **MEDIAL:** M059 (fish), M087, M067, M065, M336, M072, M089
4. **TERMINAL:** M342, M211, M162, M169, M015, M254, M012
5. **POST-TERMINAL / END:** M176, M001

Share of held-out texts whose decoded path follows the slot order:

Held-out texts	Split 1	Split 2	Split 3	Mean
Real	0.88	0.92	0.60	0.80
Shuffled	0.59	0.66	0.38	0.54
Reversed	0.46	0.69	0.29	0.48

Real texts conform more than shuffled texts in every split. Splits vary because the state boundaries between medial and pre-terminal are not stable. We therefore report the template as a **robust coarse order (initial < post-initial < medial < terminal < post-terminal)**, not as a fixed grammar.

One consequence: M099 (“two short strokes”) behaves mainly as a post-initial element bound to M267 and M391. It does not behave like a free numeral. We flag this without interpreting it.

4.4 Script type and linguistic status

Inventory.

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Corpus	Tokens	Observed types	Hapax	Chao1	Types covering 80% / 90% of tokens
M77	13,372	417	112	547	66 / 110
ICIT	18,050	712	219	939	78 / 158

ICIT counts variants separately. Mean complete-line length is 3.98 signs, and 79% of lines have ≤ 5 signs.

An inventory of several hundred signs is far above the "few dozen" typical of alphabets and syllabaries (Rao 2010). The Knorozov group made the same argument from a count of 315 signs to infer a logo-syllabic script (as reported by Parpola 1994: 60). We agree that the inventory is **incompatible with a pure alphabet or syllabary *if* the system is phonetic writing**. It does not show that the system is writing.

Status measures (Fig. 4; `linguistic_measures.csv`):

System	H(next|prev)/shuffled	r/R	r/R after within-corpus shuffle	Cr	Lee tree
Indus M77 complete lines	0.70	0.66	0.41	7.96	"linguistic: logographic"
Indus M77 de-duplicated	0.75	0.65	0.37	8.86	"
Indus ICIT complete lines	0.74	0.26	0.41	6.51	"
Indus M77 seals only	0.73	0.63	0.37	8.10	"
Tamil aksharas, length-matched	0.83	0.10	0.38	11.9	"
Sanskrit aksharas, length-matched	0.88	0.18	0.38	7.85	"
Finnish letters, length-matched	0.93	0.45	0.42	12.2	"
English letters, length-matched	0.92	0.24	0.42	11.4	"
English / Finnish / Tamil / Sanskrit words, length-matched	0.94 / 0.87 / 0.82 / 0.88	0.07 / 0.14 / 0.02 / 0.22	0.36–0.42	5.0–6.4	"
Tamil aksharas, natural sentences	0.71	0.009	0.05	13.2	"
Memoryless Zipf (Indus unigrams)	1.00	0.41	0.41	15.7	" (misclassified)
No-repeat "badges"	1.00	—	0.39	16.0	" (misclassified)
Rigid successor	0.36	—	0.38	2.56	non-linguistic

What the table shows:

1. **Ordering constraint.** Indus sign order is more constrained than any length-matched linguistic comparandum (0.70–0.75 vs 0.82–0.94), and about as constrained as Tamil aksharas in full sentences (0.71). Sproat (2014) showed that conditional-entropy measures do not separate writing from known non-linguistic systems. We could not include his non-linguistic corpora, so we draw no status conclusion from this measure.
2. **Repetition ratio.** Sproat's r/R depends strongly on text length, as he noted (r = −0.49 with mean length). At Indus lengths, every system's within-corpus shuffle gives r/R ≈ 0.36–0.42, far above the 0.10 boundary. Five of eight length-matched linguistic comparanda exceed the boundary, so it does not transfer to texts of about four signs. Sproat's own figure for "Indus bar seals" (0.58) is close to our M77 value (0.66). The ICIT transcription, however, gives 0.26, below its own shuffle. We think the difference probably reflects

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how the two transcriptions code doubled or paired signs (not verified). **The measure is not robust to the choice of transcription.**

3. **Lee tree.** It labels both memoryless and no-repeat baselines "linguistic", so it carries no information in this setting. In our implementation, with boundary-padded bigrams and 9k-token samples, even English letters come out "logographic", so the Ur thresholds do not transfer.

Conclusion: undetermined. Internal statistics at the available text lengths cannot establish whether the Indus inscriptions encode language. We assign no probability to "linguistic".

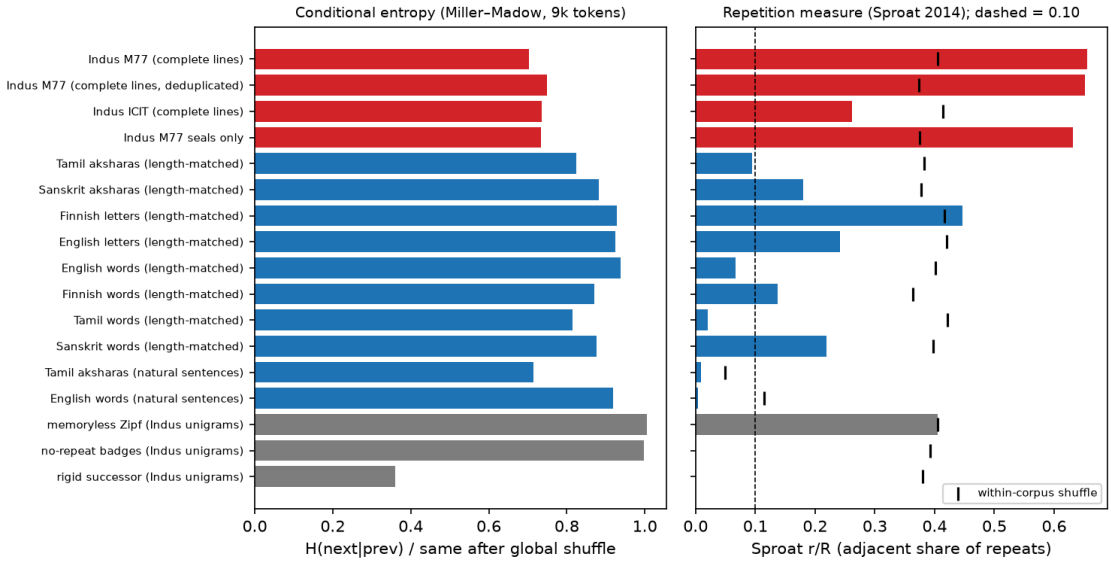


Fig. 4. Conditional-entropy ratio and Sproat's r/R for Indus corpora (red), length-matched linguistic comparanda (blue) and synthetic non-linguistic baselines (grey).

4.5 Numerals and the tablet formula

Plate check. The stroke signs match the stroke counts used in our mapping:

- Long strokes: M086–M096 = 1–5.
- Short strokes: M097–M121 = 1–12, including stacked arrangements 2/1, 2/2, 3/2, 3/3, 4/3, 4/4, 5/4 and 5/5.

Frequencies.

- Short-stroke values: 1:214, 2:852, 3:259, 4:107, 5:57, 6:56, 7:88. The values 2 (×2.2) and 7 (×4.1) are over-represented relative to a log-linear trend.
- Powers of two are not over-represented.

The W700 formula.

- On moulded tablets, W700 follows long-stroke numerals 2, 3 and 4 as 124, 186 and 130 tokens (χ^2 vs uniform $p = 0.0003$; Fig. 2). 437 of the 578 W700 tokens are at Harappa.
- The weight series (Hemmy unit) is binary up to ×64, apart from a minor 8/3 class. So the most common tablet value, 3, has no counterpart there.
- A direct "number of weight units" reading is **not supported**. This is a weak test, since the numerals could count something other than weights.

Parpola's fish claim. Parpola (1994: 194) states that numbers before the fish "are restricted to 3, 4, 6 and 7". In M77 the plate-checked short "2" (M099/M100) precedes the plain fish M059 16 times, and long "2" (M087) 67 times. The claim is contradicted as stated.

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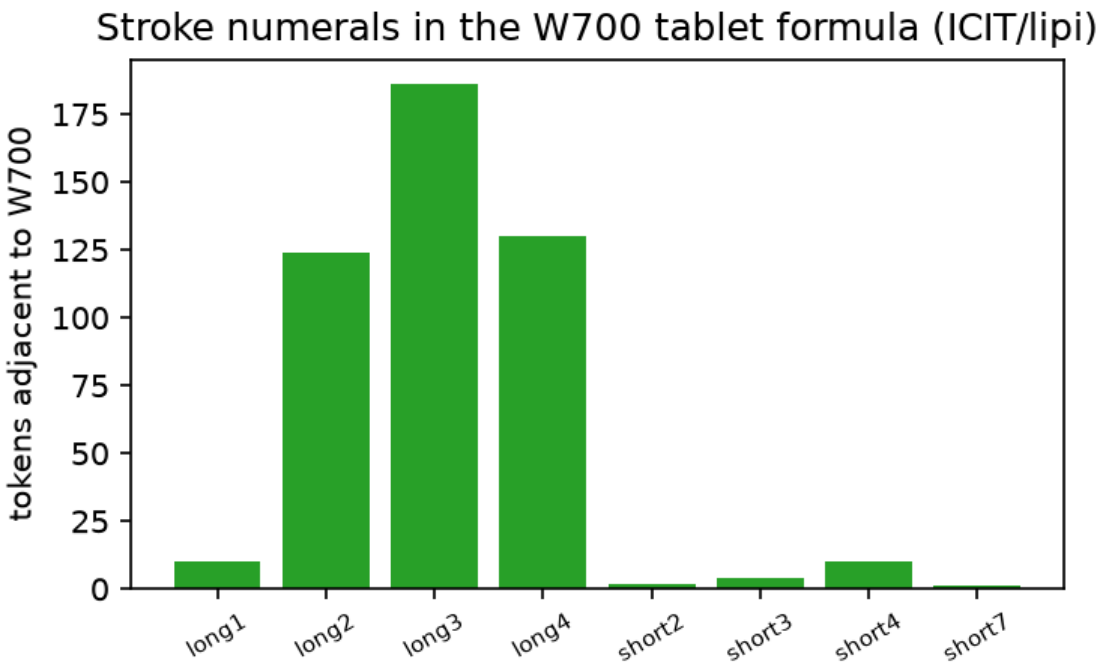


Fig. 2. Numerals adjacent to W700 (ICIT-derived).

4.6 Published phonetic readings do not beat null models

Yajnadevam (Sanskrit).

- Published readings follow from the key in 59.8% of cases by exact consonant skeleton, and in 85.9% allowing s~h and n~m.
- **B (fixed key):** real texts segment into Monier-Williams headwords at 99.35%. Within-text shuffles give 99.08% and a permuted key 99.02%.
- **C (fitted keys):** keys fitted to any of 8 lexicons reach ≥ 85% held-out "readability" on shuffled texts too. Fitting to Markov nonsense reaches 0.98.
- **E (full parse):** vidyut parses 57.9% of mechanical outputs in real order, against 59.3% shuffled.
- No test favours real sign order or the published key over controls (Fig. 3).

Dravidian rebus keys.

- The held-out compound hit rate for the Dravidian lexicon is 0.073, against 0.072 within-shuffled and 0.063 globally shuffled.
- This is lower than for Finnish (0.197), Sanskrit (0.190) or Markov nonsense (0.107). Those lexicons have higher base rates of short attested strings.
- Of Parpola's asterism compounds, only mū-m-mīn is attested in the DEDR forms; aru-mīn and e̲lu-mīn are not.
- These tests have low power against a specific, correct rebus system. They do **not** refute a Dravidian hypothesis. They show that no lexicon is singled out by the data.

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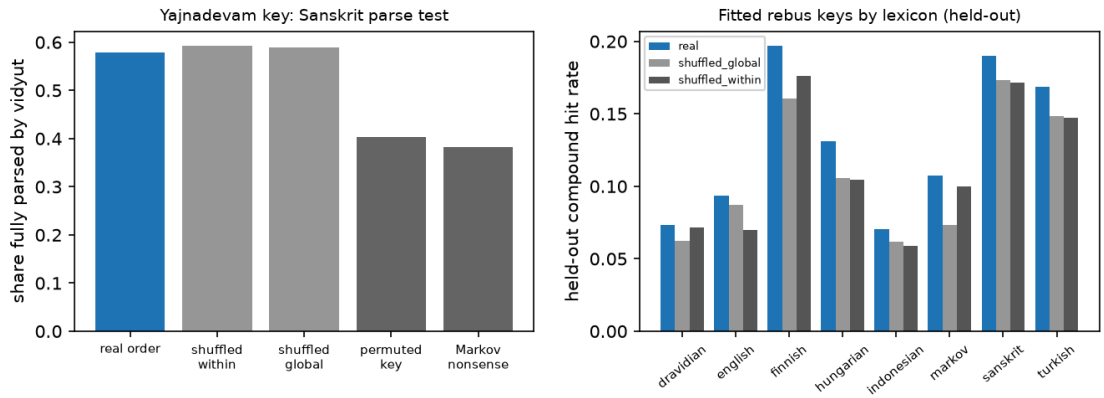


Fig. 3. Left: vidyut full-parse rate of the Yajnadevam key output. Right: held-out compound hit rate of fitted rebus keys by lexicon.

4.7 Other anchors

- **Motif:** no sign–motif association survives correction among 1,472 seals.
- **Site:** Mohenjo-daro vs Harappa differences vanish when the comparison is restricted to seals.
- **Gulf:** West-Asian/Gulf inscriptions use Indus signs in non-Indus sequences. Only 35% of their bigrams are attested in the Indus corpus, against 80% (95% interval 68–91%) for matched Indus texts. This is consistent with Laursen (2010) and with a different language or usage.
- **Tablets:** moulded tablets are mass-produced formula sets; 73–85% of them are duplicates.

4.8 Full sign map and an improved concordance

Coverage. `outputs/signmap/sign_map_full_public.csv` holds 418 M rows (417 signs plus the damage marker) and 714 W rows (712 attested). Each row gives:

- cross-scheme IDs and allographs
- frequency and positional profile, with the session-3 class (135 M signs) or a provisional binomial class
- neighbours, sites and object types
- glyph references (private)
- best foreign look-alike
- published claims and their test results

Of the 417 M signs:

- 334 have a Carlson W link, and 216 of these are model-strong or supported.
- For 48 unlinked M signs with frequency ≥ 5 the model proposes W candidates.
- Provisional classes: 50 INITIAL-like, 25 TERMINAL-like, 139 MEDIAL/unspecified, and 203 unresolved ($n < 5$).

Concordance accuracy. Under cross-validation on the 118 anchors, the combined model recovers the hand match:

Model	Top-1	Top-5
Combined (distribution + visual)	**84.7%**	94.1%
Distribution only	66.1%	—
Visual only	49.2%	—

- Chance top-1 is 0.25%.
- Per-fold Hungarian assignment reaches 85.6%.

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- A global Hungarian assignment agrees with Carlson for 73.5% of 257 linked M signs.

Link labels for the 566 Carlson links:

Label	Links
Strong	198
Supported (top 5)	54
Weak	33
Not confirmed	133
Rare, visual-only support	55
Rare, weak visual support	24
Rare, untestable	62
No glyph	7

Visual check by the AI assistant (Fig. 7).

- In the AI assistant's judgement (not a human's), 37 of 40 links look correct or plausible variants: 17/20 of the lowest-scored and 20/20 of the random sample.
- All 15 links the model supports were correct.
- 22 of the 25 links it does not confirm were also correct.
- So "not confirmed" usually reflects numerals, or a W counterpart split into several variants. It is not evidence of error.
- Doubtful links: M053/W193, M067/W242, and M112/W007 (crop clipped).

Fig. 7 (visual check of 40 concordance links, M77 crops beside SK-font renders; the judgements were made by the project's AI research assistant, not by a human) is omitted from the public version: both image sources are under copyright. The per-pair judgements are in `outputs/signmap/handcheck_pairs.csv`.

4.9 Published sign values: claims table and leave-one-out tests

Claims compiled (`outputs/claims/all_proposals.csv`):

- 42 hand-sourced rows (Appendix B), of which 28 have a sign mapping of high or medium confidence
- 714 Yajnadevam values
- 99 Field-Cady values

Phonetic keys.

- **Yajnadevam / Sanskrit.** 10 of 226 signs reach $p < .05$, where 10.5 are expected by chance; 0 survive BH correction.
- **Field-Cady / proto-Dravidian.** 4 of 99 signs reach $p < .05$; 0 survive.
- **No sign beats all nulls**, i.e. significant against alternative values, above both shuffles, and above every alternative lexicon.
- **Yajnadevam vs calibration keys.** Yajnadevam's mean per-sign p is 0.386, better than the 0.5 expected by chance. A key hill-climbed to a shuffled corpus reaches 0.406, and a key fitted from a random start reaches 0.403 (Mann–Whitney $p = .32$). The advantage is what in-sample fitting alone produces.
- **Field-Cady vs chance.** Field-Cady's mean p is 0.56.
- **Power.** The alternative inventories hold 129 and 100 values, so a single true value could not reach significance after FDR correction alone. The global comparison is the informative test.

Hand-sourced claims, site-held-out. Held-out data reproduce these predictions (BH $q < .05$):

- M267 initial (86%) and M150 initial (4/4)

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- M342 final (74%), which supports Parpola's ā, Knorozov's -at(tu) and Mahadevan's -(a)nṟu equally
- M211 final (38/39)
- M267+M099 as a unit (56 vs 15.9 expected)
- 6+fish (3 vs 0.63), 3+fish (3 vs 0.61) and M087+fish (7 vs 1.75)
- M403+M087 (4 vs 0.98)

Held-out data do **not** support these:

- M284, M264 and M149 as initial. M149 is not initial even in-sample: 11% against a 25% null.
- M261/M373 as initial
- 7+fish (no held-out text contains both signs; in-sample 1 adjacency vs 1.39 expected)
- Mahadevan's M342/M211 gender paradigm (left-context similarity p = .32)
- Wells' "cow at the city" (M267+M342 adjacency 3 vs 5.9 expected; the two signs sit at opposite ends of texts)

The confirmed predictions confirm the **role** each claim assigns, never the sound value. Any reading that gives the jar a suffix role, or treats M267+M099 as one unit, passes the same tests.

4.10 Branching-entropy segmentation and NSB entropy

Branching entropy vs BPE.

- Branching-entropy and BPE boundaries agree beyond chance on held-out seal texts: F1 0.49 against 0.42 for random boundaries at the same rate (p = .001). On shuffled corpora the agreement is 0.55 against 0.53.
- Recurring branching-entropy units match the BPE units: M267+M099 (143), M391+M099 (52), M150+M123 (24), M293+M123+M343 (17), M087+M059 (16), M162+M342 (15).

HMM boundaries. The HMM state changes at 74% of sign junctions. It agrees with neither branching entropy (F1 0.41 vs 0.43) nor BPE. Its states are per-sign position slots, not units.

NSB entropy (bits):

Corpus	Unigram entropy	Conditional, real	Conditional, shuffled
M77	6.754 ± 0.005	4.29	5.65
ICIT	7.14 ± 0.006	4.62	5.84

- The M77 unigram figure is insensitive to alphabet size: 6.753 to 6.756 for K = 398, 417 or Chao1 521.
- The order effect is robust to the estimator: NSB, Miller–Madow (3.88) and ML (3.70) agree on it.

4.11 Comparative look-alikes predict nothing beyond strokes

Category tests.

Script and category	p, all signs	p, strokes excluded
Egyptian Gardiner group	.020 (MI 0.663 vs 0.553)	.57
Linear A category	.019	.20
Brahmi	.47	.36
Cuneiform numeric/sign	.38	.75

- No test survives BH correction across the eight tests (minimum q ≈ .08).
- The only residue is that strokes resemble strokes everywhere.

Indus-internal similarity. Visual similarity does not predict positional similarity (Mantel ρ = 0.009, p = .40; 143 signs).

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Match quality. Best-match scores are no higher than against equally sized random glyph sets from other scripts. Most of the highest-scoring look-alikes are stroke signs (Fig. 8). Nothing here supports a genetic link between the scripts.

Fig. 8 (top visual look-alikes, M77 crops beside foreign glyphs) is omitted from the public version because the M77 crops are under copyright. Scores are in `outputs/comparanda/lookalikes_top3.csv`.

4.12 Prior results replicated, and a synthetic emblem system passes our structural tests

Table 12a. Held-out perplexity by n-gram order (5-fold CV, Witten–Bell).

Corpus	n=1	n=2	n=3	n=4	n=5
Yadav et al. 2010, EBUDS (published)	68.82	26.69	26.09	25.26	25.26
M77 EBUDS-like, dedup (ours)	74.84	31.33	32.75	38.54	45.47
M77 with duplicates	66.76	21.29	18.63	20.01	22.25
ICIT dedup	89.90	42.04	44.66	52.67	62.48
M77 dedup, within-text shuffled	74.84	79.81	99.64	116.31	135.14
SIGIL-like synthetic (non-linguistic)	104.31	36.14	43.12	51.37	60.56

- The large bigram gain replicates in both corpora.
- On deduplicated texts, orders above 2 do not help in our implementation; they help only when duplicates are kept. Yadav's trigram and quadrigram gains are small (26.69 → 25.26), so differences in corpus filtering and smoothing details could account for the discrepancy. We report it as **not replicated** rather than refuted.
- The synthetic non-linguistic system shows the same bigram drop.

Table 12b. Restoring one dropped sign (5-fold CV).

Corpus	Scorer	Top-1	Top-5	Sensitivity (90% set)	Mean 90%-set size
M77 EBUDS-like dedup (n = 1,761)	frequency only	0.116	0.250	0.878	120
	position + frequency	0.161	0.328	0.861	96
	WB bigram	0.344	0.593	0.812	37
	WB trigram	0.357	0.589	0.738	24
ICIT dedup (n = 2,159)	frequency only	0.101	0.200	0.875	186
	WB bigram	0.285	0.514	0.792	61
SIGIL-like (n = 1,753)	frequency only	0.041	0.163	0.884	182
	WB bigram	0.219	0.461	0.736	33

Yadav et al. report 74% sensitivity. A frequency-only scorer exceeds that, because its 90% set is large. Sensitivity without set size is therefore uninformative. Top-1 shows a genuine sequential gain: about 3× over frequency in both corpora. That gain is also present in the synthetic system.

Nair (2026) replication.

Corpus	Texts	Signs	Tokens	Zipf slope (R^2)	H_cond, bits	Permutation mean ± sd	p
Nair 2026 (published)	1,916	584	11,110	−1.49 (0.96)	3.232	n/s	percentile 0 of 1,000

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Corpus	Texts	Signs	Tokens	Zipf slope (R^2)	H_cond, bits	Permutation mean ± sd	p
ICIT/lipi dedup, undamaged	2,273	664	9,662	−1.40 (0.96)	3.433	4.616 ± 0.015	0.001
ICIT/lipi dedup, all segments	3,225	712	12,377	−1.44 (0.96)	3.604	4.774 ± 0.015	0.001
M77 EBUDS-like dedup	1,849	398	8,313	−1.52 (0.93)	3.346	4.663 ± 0.015	0.001
SIGIL-like synthetic	1,808	356	—	−1.18 (0.86)	3.541	4.508 ± 0.013	0.003 (300 perms)

Both values replicate within the range expected from different deduplication and damage filters. The plug-in H_cond depends on corpus size, so its absolute value should not be compared across corpora. The permutation gap is just as large in the synthetic emblem system.

HMM slot grammar on the synthetic system. Held-out log-likelihood gain from K = 1 to K = 5 states (nats per token):

- M77 dedup: 0.44 real vs 0.003 shuffled.
- SIGIL-like: 0.93 real vs −0.18 shuffled.

A field-structured emblem system thus produces a *stronger* slot signal than the Indus texts. Our slot grammar (§4.3) is evidence of ordered fields, and it is equally compatible with an administrative or emblematic code.

Fuls-style positions. For 124 M77 signs with ≥ 10 tokens:

- 93% of our INITIAL signs fall in the initial or early-medial bins.
- 73% of our TERMINAL signs fall in the terminal bin.
- 53% of our PRE-TERMINAL signs fall in the late-medial or terminal bins.

Our classes broadly agree with a Fuls-style normalised position (crosstab in `outputs/compare/fuls_vs_our_classes_crosstab.csv`). The pre-terminal class is defined by adjacency to terminals, not by mean position, so the lower agreement is expected.

Verdict.

- **Stronger than prior work:** calibration (FDR, bootstrap), held-out and site-held-out design, cross-corpus replication, and per-sign tests of published values.
- **Weaker:** corpus quality (no access to full ICIT), real non-linguistic comparanda, and breadth of the specificity registry.
- **Shared with all prior work:** the structural measures detect organisation, not language.

5. Sign-by-sign assessment

Appendix A lists the 100 most frequent M77 signs plus all stroke numerals: 116 rows, with full columns in `outputs/structure/sign_table.csv`. Confidence is assigned at three levels.

1. **Positional/structural function: calibrated percentage.**
    - Thirty-eight signs reach ≥ 85%.
    - Of these, 28 also replicate in ICIT:
        - **TERMINAL:** M342, M176, M328, M169, M211, M001, M015, M012, M254, M060, M242, M139
        - **PRE-TERMINAL:** M048, M347, M008, M124, M296, M135, M326, M358, M193, M062
        - **INITIAL:** M267, M391, M204, M150, M177, M041

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    - The percentage is the probability that the sign belongs to the stated positional class, given the corpus and the null model. It is not a probability for any meaning.
2. **Numeral function: qualitative "high".** It rests on iconicity checked on the plates and on substitution in formulae. Distributional cohesion fails, so no percentage is given.
3. **Phonetic/semantic value: "not estimable above null"** for every proposal we could test (Yajnadevam, Parpola, Mahadevan). The status of proposals we could not test is "no proposal tested".

Of the 116 rows, none carries a phonetic value at any confidence above null.

Session 4 extension. The full map is `outputs/signmap/sign_map_full_public.csv` (1,132 rows). It also lives in the SQLite table `sign_map_full`.

- Signs outside the session-3 class set receive a *provisional* class from binomial tests against the base rates: 50 INITIAL-like and 25 TERMINAL-like M signs.
- These provisional classes carry p-values but no calibrated percentage.
- Appendix B lists the hand-sourced claims with their held-out outcomes.

6. Limitations

- **Corpora.** Both corpora are secondary transcriptions. The M ↔ W concordance is unverified and many-to-many, and we used it for replication. ICIT-derived data has unclear provenance.
- **Text length.** Short texts (mean ≈ 4) and duplication limit every statistical test. The non-linguistic status tests lack Sproat's real non-linguistic corpora.
- **Entropy estimation.** Block entropies used Miller–Madow. NSB estimates were added in session 4 (§4.10); NSB overestimates entropy in sparse contexts, so absolute values differ between estimators while the real-vs-shuffled gap is robust.
- **Search depth.** Fitted-key searches are short hill-climbs (2,000 iterations for the segmentability keys, 4,000 for the rebus keys), so they underestimate what an optimiser can reach. Null results on "readability" are therefore conservative. Positive phonetic results would need far stronger controls.
- **Lexicons.** The reduced 3-vowel alphabet used for cross-lexicon comparison is coarse. The DEDR forms lack many compounds.
- **Not done:** re-OCR of the remaining 305 M77 object codes (10.5%). Branching-entropy segmentation and per-sign leave-one-out tests were added in session 4.
- **Session-4 limits.**
    - The sign crops come from a 150-dpi scan, and M049 is missing.
    - The W renders come from a modern font, not from the inscriptions.
    - Per-sign lexical tests have low power: the alternative-value inventories hold 100–129 values.
    - Mappings of signs that appear only in figures were made by glyph similarity and carry the stated confidence.
    - Parpola's Fig. 15.2 nos. 1–10 (p. 275) are missing from the scan, so they are not in the claims table.
    - Proto-Elamite, archaic proto-cuneiform and South Indian graffiti marks have no open glyph source and were not compared.
- **Session-5 limits.**
    - Replications use our own implementations; Yadav's smoothing details and Nair's exact corpus filters may differ from ours.
    - Fuls' positional formula is approximated from an abstract.
    - Our synthetic control is one hand-designed generator, not Raghavendra's registered 54-method battery.
    - Koskenniemi (1981), Fuls (2013), FLAIR and Tamkiya et al. were seen only as abstracts or second-hand.

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- **AI-made visual checks.** Every visual judgement in this paper (the 40-link concordance check, the plate checks of numerals and fish signs, the NUMERAL-STROKE tags and glyph-crop spot checks) was made by the project's AI research assistant, not by a human epigrapher. None has yet been independently checked by a person. A blind human protocol is planned.
- **Run-to-run variation in controls (found in session 6).** The Markov-nonsense control lexicon and the fitted rebus keys depend on Python's per-process string-hash order, so these control values shift between runs (fitted rebus scores by up to about 0.02; Markov-nonsense fixed-key segmentability between 0.76 and 0.91). The published-key results do not change, and no qualitative conclusion changed in a fresh-clone rerun. A deterministic rerun with reported spread is planned.
- **Scope of the original brief.** The request for "85% confidence in a complete decipherment" cannot be met. Without a bilingual or an independent semantic anchor, no current method can calibrate phonetic values (Parpola 1994: 48).

7. Data and code availability

This public version ships with code (MIT), our text and derived result tables (CC BY 4.0), and a small set of openly licensed images. It does **not** include the input corpora:

- the M77 export: ASI copyright, extracted via a logged-in session;
- the *lipi*/ICIT-derived texts and sign key: no licence;
- OCR of Mahadevan 1977 and Parpola 1994;
- plate crops and SK-font renders.

`scripts/fetch_data.sh` downloads the openly hosted inputs and explains how to obtain the rest. Users must check each upstream licence themselves. `DATA_POLICY.md` lists every exclusion and the reason for it. Figures 7 and 8 are omitted. Yajnadevam/*lipi* key values are redacted from the tables; the per-sign test statistics are kept.

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Supplementary figures

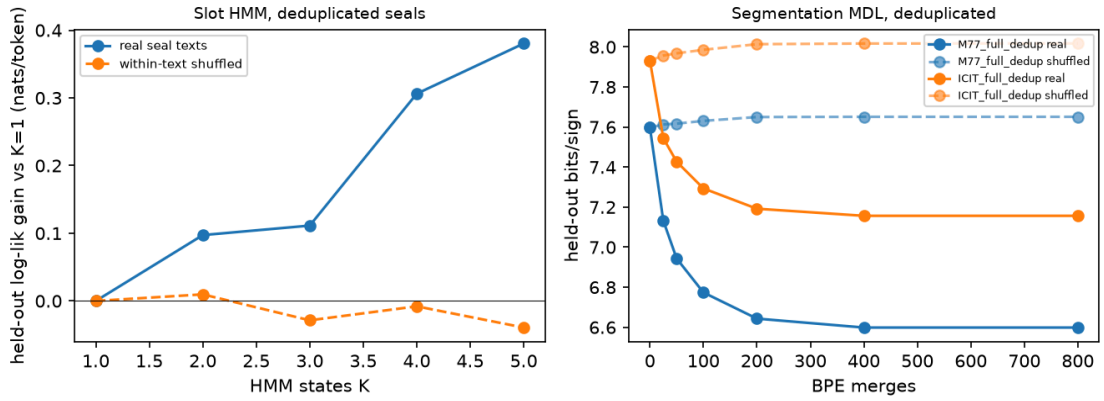


Fig. 5. Left: held-out HMM gain on de-duplicated seal texts, real vs shuffled. Right: segmentation description length after de-duplication.

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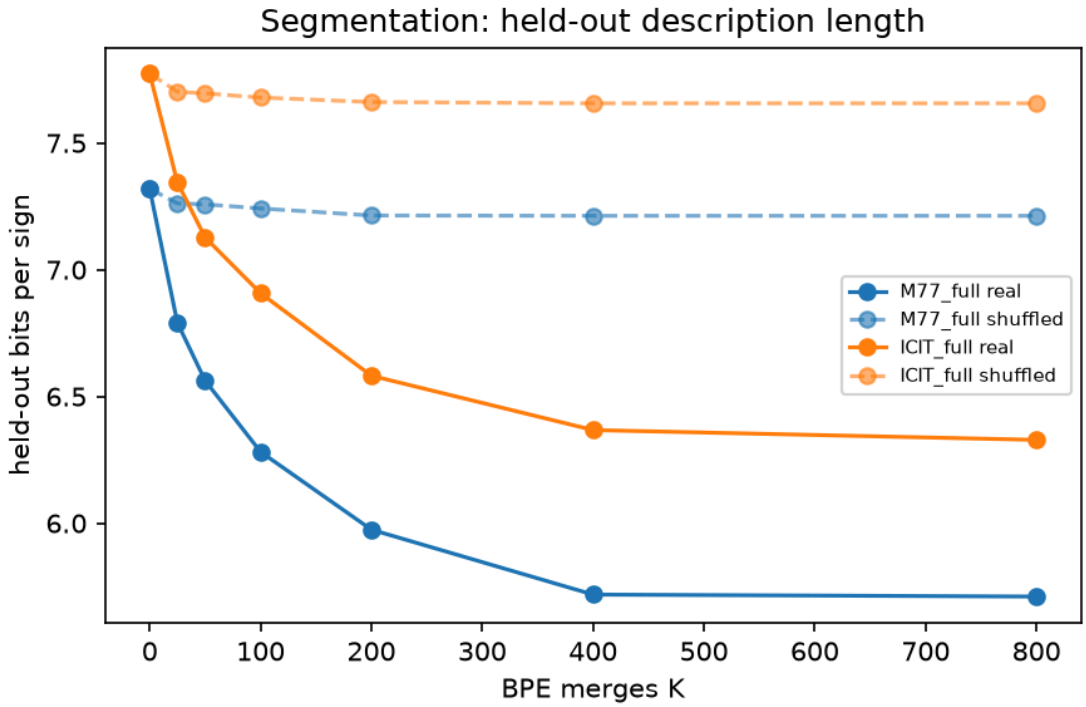


Fig. S1. Segmentation description length before de-duplication.

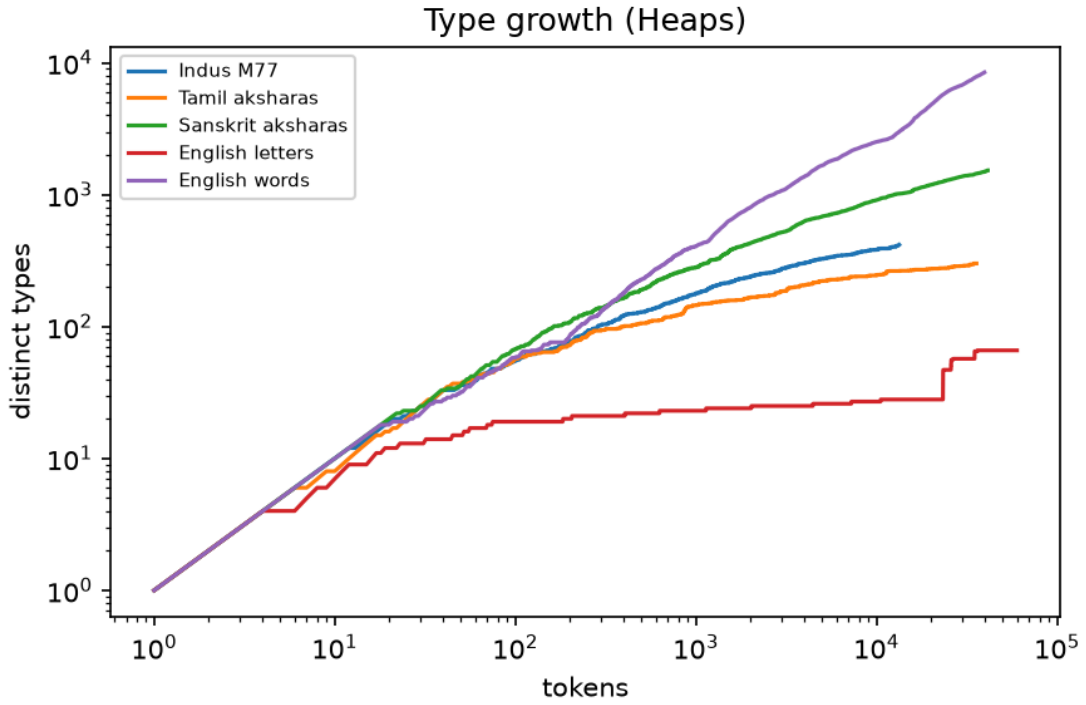


Fig. S2. Type growth (Heaps curves): Indus vs comparanda.

Appendix A. Sign table (top 100 M77 signs + all stroke numerals)

Columns: M = Mahadevan 1977; P = Parpola; W = Wells/ICIT (via Carlson's unverified concordance); freq = M77 tokens; init/final share = among multi-sign lines. "function_confidence" is calibrated only for positional classes (see §3.1). "phonetic_value_confidence" reports the outcome of the null-model tests in §4.6.

M	P	W	freq	init_share	final_share	structural_class	literature_values	our_function	function_confidence	phonetic_value_confidence
M342	P324	W740	1395	0.015	0.702	TERMINAL	Yajnadevam/lipi syllabic key	text-final marker	99.8% (positional class)	not estimable above null

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M	P	W	freq	init_share	final_share	structural_class	literature_values	our_function	function_confidence	phonetic_value_confidence
							value (not redistributed) | jar; most frequent end sign — Mahadevan 2015	(suffix/determinative/title-like); semantic value unknown		
M099	P119; P122	W002	649	0.039	0.076	NUMERAL-STROKE	Yajnadevam/lipi syllabic key value (not redistributed) | numeral (stroke count) — Mahadevan 1977 sign list grouping; Parpola 1994 (e.g. p. 194)	numeral, value 2 (short strokes)	qualitative: high	not estimable above null
M059	P050	W220	381	0.14	0.095	MEDIAL/UNSPECIFIED	Yajnadevam/lipi syllabic key value (not redistributed) | fish = Dravidian *mīn " fish/star" (rebus) — Parpola 1994	no distinctive positional behaviour (medial/unspecified)	—	not estimable above null
M267	P385	W817; W861	376	0.797	0.046	INITIAL	Yajnadevam/lipi syllabic key value (not redistributed) | fortified house (place sign) — Mahadevan 2015	text-initial sign (title/place/category-like); semantic value unknown	99.7% (positional class)	not estimable above null
M087	P145	W032	365	0.409	0.059	NUMERAL-STROKE	Yajnadevam/lipi syllabic key value (not redistributed) | numeral (stroke count) — Mahadevan 1977 sign list grouping; Parpola 1994 (e.g. p. 194)	numeral, value 2 (long strokes)	qualitative: high	not estimable above null
M176	P098	W400; W402	355	0.0	0.891	TERMINAL	Yajnadevam/lipi syllabic key value (not redistributed)	text-final marker (suffix/determinative/title-like); semantic value unknown	99.8% (positional class)	not estimable above null
M328	P310	W700	323	0.06	0.851	TERMINAL	Yajnadevam/lipi syllabic key value (not redistributed)	text-final marker (suffix/determinative/title-like); semantic value unknown	99.8% (positional class)	not estimable above null
M089	P147	W033	314	0.444	0.069	NUMERAL-STROKE	Yajnadevam/lipi syllabic key value (not redistributed) | numeral (stroke count) — Mahadevan 1977 sign list grouping; Parpola 1994 (e.g. p. 194)	numeral, value 3 (long strokes)	qualitative: high	not estimable above null
M067	P062	W240; W242	279	0.272	0.047	MEDIAL/UNSPECIFIED		no distinctive positional behaviour (medial/unspecified)	—	no proposal tested
M169	P086	W390; W392; W405;	240	0.021	0.565	TERMINAL	Yajnadevam/lipi syllabic key value (not redistributed)	text-final marker (suffix/determinative/title	99.8% (positional class)	not estimable above null

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M	P	W	freq	init_share	final_share	structural_class	literature_values	our_function	function_confidence	phonetic_value_confidence
		W406; W407; W408						-like); semantic value unknown		
M336	P316	W704; W705; W706; W707	236	0.319	0.034	MEDIAL/UNSPECIFIED	Yajnadevam/lipi syllabic key value (not redistributed)	no distinctive positional behaviour (medial/unspecified)	—	not estimable above null
M211	P217	W520	227	0.0	0.821	TERMINAL	Yajnadevam/lipi syllabic key value (not redistributed)	text-final marker (suffix/determinative/title-like); semantic value unknown	99.8% (positional class)	not estimable above null
M065	P060	W235	216	0.395	0.051	MEDIAL/UNSPECIFIED		no distinctive positional behaviour (medial/unspecified)	—	no proposal tested
M162	P086	W390; W392; W405; W406; W407; W408	212	0.057	0.459	MEDIAL/UNSPECIFIED	Yajnadevam/lipi syllabic key value (not redistributed)	no distinctive positional behaviour (medial/unspecified)	—	not estimable above null
M245	P268	W615	207	0.137	0.239	MEDIAL/UNSPECIFIED	Yajnadevam/lipi syllabic key value (not redistributed)	no distinctive positional behaviour (medial/unspecified)	—	not estimable above null
M391	P378	W820	195	0.706	0.086	INITIAL	Yajnadevam/lipi syllabic key value (not redistributed)	text-initial sign (title/place/category-like); semantic value unknown	99.7% (positional class)	not estimable above null
M123	P120	W060	193	0.021	0.062	MEDIAL/UNSPECIFIED	Yajnadevam/lipi syllabic key value (not redistributed)	no distinctive positional behaviour (medial/unspecified)	—	not estimable above null
M072	P058	W233	188	0.237	0.043	MEDIAL/UNSPECIFIED		no distinctive positional behaviour (medial/unspecified)	—	no proposal tested
M343	P325	W741	177	0.137	0.063	MEDIAL/UNSPECIFIED	Yajnadevam/lipi syllabic key value (not redistributed)	no distinctive positional behaviour (medial/unspecified)	—	not estimable above null
M249	P276	W590	170	0.25	0.095	MEDIAL/UNSPECIFIED	Yajnadevam/lipi syllabic key value (not redistributed)	no distinctive positional behaviour (medial/unspecified)	—	not estimable above null
M048	P194	W176; W177; W178; W179	168	0.455	0.042	PRE-TERMINAL	Yajnadevam/lipi syllabic key value (not redistributed)	occurs immediately before terminal signs (modifier/inflection-like slot)	89.0% (positional class)	not estimable above null
M102	P123	W003	151	0.503	0.049	NUMERAL-STROKE	Yajnadevam/lipi syllabic key value (not redistributed) | ' 3' +fish = mū-m-mīn = Rampla 1994 | numeral (stroke count) —	numeral, value 3 (short strokes)	qualitative: high	not estimable above null

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M	P	W	freq	init_share	final_share	structural_class	literature_values	our_function	function_confidence	phonetic_value_confidence
							Mahadevan 1977 sign list grouping; Parpola 1994 (e.g. p. 194)			
M086	P144	W031	149	0.463	0.109	NUMERAL-STROKE	Yajnadevam/lipi syllabic key value (not redistributed) | numeral (stroke count) — Mahadevan 1977 sign list grouping; Parpola 1994 (e.g. p. 194)	numeral, value 1 (long strokes)	qualitative: high	not estimable above null
M293	P160	W920	136	0.537	0.044	INITIAL	Yajnadevam/lipi syllabic key value (not redistributed)	text-initial sign (title/place/category-like); semantic value unknown	83.7% (positional class)	not estimable above null
M389			134	0.492	0.023	MEDIAL/UNSPECIFIED		no distinctive positional behaviour (medial/unspecified)	—	no proposal tested
M001	P013	W090	134	0.123	0.692	TERMINAL	Yajnadevam/lipi syllabic key value (not redistributed)	text-final marker (suffix/determinative/title-like); semantic value unknown	99.8% (positional class)	not estimable above null
M171	P092	W415	132	0.333	0.045	MEDIAL/UNSPECIFIED	Yajnadevam/lipi syllabic key value (not redistributed)	no distinctive positional behaviour (medial/unspecified)	—	not estimable above null
M053	P075	W193; W797; W798	130	0.295	0.054	MEDIAL/UNSPECIFIED	Yajnadevam/lipi syllabic key value (not redistributed)	no distinctive positional behaviour (medial/unspecified)	—	not estimable above null
M015	P004	W153; W154; W155; W156; W158	126	0.017	0.742	TERMINAL	Yajnadevam/lipi syllabic key value (not redistributed)	text-final marker (suffix/determinative/title-like); semantic value unknown	99.8% (positional class)	not estimable above null
M347	P332	W752; W760	118	0.077	0.034	PRE-TERMINAL	Yajnadevam/lipi syllabic key value (not redistributed)	occurs immediately before terminal signs (modifier/inflection-like slot)	100.0% (positional class)	not estimable above null
M008	P009	W100	105	0.048	0.048	PRE-TERMINAL	Yajnadevam/lipi syllabic key value (not redistributed)	occurs immediately before terminal signs (modifier/inflection-like slot)	100.0% (positional class)	not estimable above null
M051	P073	W255	105	0.272	0.068	MEDIAL/UNSPECIFIED	Yajnadevam/lipi syllabic key value (not redistributed)	no distinctive positional behaviour (medial/unspecified)	—	not estimable above null
M387	P364	W803; W805; W806	102	0.34	0.05	MEDIAL/UNSPECIFIED	Yajnadevam/lipi syllabic key	no distinctive positional	—	not estimable above null

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M	P	W	freq	init_share	final_share	structural_class	literature_values	our_function	function_confidence	phonetic_value_confidence
							value (not redistributed)	behaviour (medial/unspecified)		
M402	P202	W365; W367; W368	99	0.103	0.34	MEDIAL/UNSPECIFIED	Yajnadevam/lipi syllabic key value (not redistributed)	no distinctive positional behaviour (medial/unspecified)	—	not estimable above null
M403	P349	W840	93	0.189	0.044	MEDIAL/UNSPECIFIED	Yajnadevam/lipi syllabic key value (not redistributed)	no distinctive positional behaviour (medial/unspecified)	—	not estimable above null
M149	P255	W690	92	0.124	0.247	MEDIAL/UNSPECIFIED	Yajnadevam/lipi syllabic key value (not redistributed)	no distinctive positional behaviour (medial/unspecified)	—	not estimable above null
M017	P011	W140	91	0.444	0.078	MEDIAL/UNSPECIFIED	Yajnadevam/lipi syllabic key value (not redistributed)	no distinctive positional behaviour (medial/unspecified)	—	not estimable above null
M097	P121	W001	91	0.033	0.066	NUMERAL-STROKE	Yajnadevam/lipi syllabic key value (not redistributed) | numeral (stroke count) — Mahadevan 1977 sign list grouping; Parpola 1994 (e.g. p. 194)	numeral, value 1 (short strokes)	qualitative: high	not estimable above null
M216	P230; P234	W550; W551; W555	90	0.453	0.105	MEDIAL/UNSPECIFIED	Yajnadevam/lipi syllabic key value (not redistributed)	no distinctive positional behaviour (medial/unspecified)	—	not estimable above null
M244	P270	W630; W631; W632; W633; W634; W635; W636; W637; W638; W639	89	0.352	0.068	MEDIAL/UNSPECIFIED	Yajnadevam/lipi syllabic key value (not redistributed)	no distinctive positional behaviour (medial/unspecified)	—	not estimable above null
M287	P154	W900	88	0.239	0.068	MEDIAL/UNSPECIFIED	Yajnadevam/lipi syllabic key value (not redistributed)	no distinctive positional behaviour (medial/unspecified)	—	not estimable above null
M098	P121	W001	88	0.045	0.148	NUMERAL-STROKE	Yajnadevam/lipi syllabic key value (not redistributed) | numeral (stroke count) — Mahadevan 1977 sign list grouping; Parpola 1994 (e.g. p. 194)	numeral, value 1 (short strokes)	qualitative: high	not estimable above null
M012	P001	W151	80	0.026	0.87	TERMINAL	Yajnadevam/lipi syllabic key value (not redistributed)	text-final marker (suffix/determinative/title-like); semantic value unknown	99.8% (positional class)	not estimable above null
M124	P204	W061; W062; W063	78	0.256	0.038	PRE-TERMINAL	Yajnadevam/lipi syllabic key value (not redistributed)	occurs immediately before terminal signs	100.0% (positional class)	not estimable above null

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M	P	W	freq	init_share	final_share	structural_class	literature_values	our_function	function_confidence	phonetic_value_confidence
								(modifier/inflection-like slot)		
M204	P210	W503	76	0.662	0.014	INITIAL	Yajnadevam/lipi syllabic key value (not redistributed)	text-initial sign (title/place/category-like); semantic value unknown	99.7% (positional class)	not estimable above null
M254	P301	W526; W527	73	0.041	0.836	TERMINAL	Yajnadevam/lipi syllabic key value (not redistributed)	text-final marker (suffix/determinative/title-like); semantic value unknown	99.8% (positional class)	not estimable above null
M070	P056	W231	73	0.247	0.096	MEDIAL/UNSPECIFIED		no distinctive positional behaviour (medial/unspecified)	—	no proposal tested
M121	P142	W055	70	0.186	0.071	NUMERAL-STROKE	Yajnadevam/lipi syllabic key value (not redistributed) | numeral (stroke count) — Mahadevan 1977 sign list grouping; Parpola 1994 (e.g. p. 194)	numeral, value 12 (short strokes)	qualitative: high	not estimable above null
M104	P124	W004; W014	70	0.441	0.044	NUMERAL-STROKE	Yajnadevam/lipi syllabic key value (not redistributed) | numeral (stroke count) — Mahadevan 1977 sign list grouping; Parpola 1994 (e.g. p. 194)	numeral, value 4 (short strokes)	qualitative: high	not estimable above null
M112	P127	W007; W017	70	0.286	0.0	NUMERAL-STROKE	Yajnadevam/lipi syllabic key value (not redistributed) | numeral (stroke count) — Mahadevan 1977 sign list grouping; Parpola 1994 (e.g. p. 194)	numeral, value 7 (short strokes)	qualitative: high	not estimable above null
M307	P182	W892	69	0.212	0.258	MEDIAL/UNSPECIFIED	Yajnadevam/lipi syllabic key value (not redistributed)	no distinctive positional behaviour (medial/unspecified)	—	not estimable above null
M095	P150	W034	64	0.918	0.016	NUMERAL-STROKE	Yajnadevam/lipi syllabic key value (not redistributed) | numeral (stroke count) — Mahadevan 1977 sign list grouping; Parpola 1994 (e.g. p. 194)	numeral, value 4 (long strokes)	qualitative: high	not estimable above null
M150	P256	W692	63	0.793	0.017	INITIAL	Yajnadevam/lipi syllabic key value (not redistributed)	text-initial sign (title/place/category-like); semantic value unknown	99.7% (positional class)	not estimable above null

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M	P	W	freq	init_share	final_share	structural_class	literature_values	our_function	function_confidence	phonetic_value_confidence
M130	P188	W435; W436	63	0.032	0.111	BOUND-COLLOCATE	Yajnadevam/lipi syllabic key value (not redistributed)	part of a fixed collocation (ligature/compound-like)	64.8% (positional class)	not estimable above null
M373	P355	W790; W850	61	0.35	0.233	MEDIAL/UNSPECIFIED	Yajnadevam/lipi syllabic key value (not redistributed) | house (place sign) — Mahadevan 2015	no distinctive positional behaviour (medial/unspecified)	—	not estimable above null
M197	P289	W575	60	0.424	0.085	MEDIAL/UNSPECIFIED	Yajnadevam/lipi syllabic key value (not redistributed)	no distinctive positional behaviour (medial/unspecified)	—	not estimable above null
M341	P320	W716; W717	59	0.254	0.068	MEDIAL/UNSPECIFIED	Yajnadevam/lipi syllabic key value (not redistributed)	no distinctive positional behaviour (medial/unspecified)	—	not estimable above null
M194	P288		58	0.088	0.07	PRE-TERMINAL		occurs immediately before terminal signs (modifier/inflection-like slot)	86.0% (positional class)	no proposal tested
M375	P366	W832	57	0.268	0.25	MEDIAL/UNSPECIFIED	Yajnadevam/lipi syllabic key value (not redistributed)	no distinctive positional behaviour (medial/unspecified)	—	not estimable above null
M175	P091	W350; W352; W353; W354	54	0.385	0.096	MEDIAL/UNSPECIFIED	Yajnadevam/lipi syllabic key value (not redistributed)	no distinctive positional behaviour (medial/unspecified)	—	not estimable above null
M230	P215	W460	54	0.057	0.151	MEDIAL/UNSPECIFIED	Yajnadevam/lipi syllabic key value (not redistributed)	no distinctive positional behaviour (medial/unspecified)	—	not estimable above null
M294	P172	W904; W927	53	0.132	0.038	PRE-TERMINAL	Yajnadevam/lipi syllabic key value (not redistributed)	occurs immediately before terminal signs (modifier/inflection-like slot)	100.0% (positional class)	not estimable above null
M025	P031	W142	53	0.231	0.096	MEDIAL/UNSPECIFIED	Yajnadevam/lipi syllabic key value (not redistributed)	no distinctive positional behaviour (medial/unspecified)	—	not estimable above null
M345	P327	W745; W746; W747	51	0.542	0.042	INITIAL	Yajnadevam/lipi syllabic key value (not redistributed)	text-initial sign (title/place/category-like); semantic value unknown	76.7% (positional class)	not estimable above null
M252	P280	W595	51	0.059	0.353	MEDIAL/UNSPECIFIED	Yajnadevam/lipi syllabic key value (not redistributed)	no distinctive positional behaviour (medial/unspecified)	—	not estimable above null
M127	P283	W440	50	0.186	0.093	MEDIAL/UNSPECIFIED	Yajnadevam/lipi syllabic key value (not redistributed)	no distinctive positional behaviour (medial/unspecified)	—	not estimable above null

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M	P	W	freq	init_share	final_share	structural_class	literature_values	our_function	function_confidence	phonetic_value_confidence
M028	P035	W125; W126	50	0.347	0.122	MEDIAL/UNSPECIFIED	Yajnadevam/lipi syllabic key value (not redistributed)	no distinctive positional behaviour (medial/unspecified)	—	not estimable above null
M155	P114	W070; W072	49	0.102	0.02	MEDIAL/UNSPECIFIED	Yajnadevam/lipi syllabic key value (not redistributed)	no distinctive positional behaviour (medial/unspecified)	—	not estimable above null
M407	P352	W845	48	0.021	0.062	PRE-TERMINAL	Yajnadevam/lipi syllabic key value (not redistributed)	occurs immediately before terminal signs (modifier/inflection-like slot)	62.0% (positional class)	not estimable above null
M180	P205	W289; W290; W291; W292; W293; W294; W295; W296; W297; W299; W300; W301; W302; W303; W304; W305; W306	44	0.116	0.186	MEDIAL/UNSPECIFIED	Yajnadevam/lipi syllabic key value (not redistributed)	no distinctive positional behaviour (medial/unspecified)	—	not estimable above null
M182	P303	W322; W323; W324; W325	43	0.349	0.093	BOUND-COLLOCATE	Yajnadevam/lipi syllabic key value (not redistributed)	part of a fixed collocation (ligature/compound-like)	90.3% (positional class)	not estimable above null
M327	P110	W388	42	0.317	0.024	PRE-TERMINAL	Yajnadevam/lipi syllabic key value (not redistributed)	occurs immediately before terminal signs (modifier/inflection-like slot)	82.0% (positional class)	not estimable above null
M284	P382	W877	41	0.45	0.3	MEDIAL/UNSPECIFIED	Yajnadevam/lipi syllabic key value (not redistributed) | city (place sign) — Mahadevan 2015	no distinctive positional behaviour (medial/unspecified)	—	not estimable above null
M109	P126	W006; W016	38	0.583	0.028	NUMERAL-STROKE	Yajnadevam/lipi syllabic key value (not redistributed) | numeral (stroke count) — Mahadevan 1977 sign list grouping; Parpola 1994 (e.g. p. 194)	numeral, value 6 (short strokes)	qualitative: high	not estimable above null
M106	P125	W005; W015	38	0.486	0.054	NUMERAL-STROKE	Yajnadevam/lipi syllabic key value (not redistributed) | numeral (stroke count) — Mahadevan 1977 sign list grouping; Parpola 1994 (e.g. p. 194)	numeral, value 5 (short strokes)	qualitative: high	not estimable above null
M173	P095	W416	38	0.622	0.135	INITIAL	Yajnadevam/lipi syllabic key	text-initial sign	84.7% (positional class)	not estimable above null

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M	P	W	freq	init_share	final_share	structural_class	literature_values	our_function	function_confidence	phonetic_value_confidence
							value (not redistributed)	(title/place/category-like); semantic value unknown		
M296	P175	W923	35	0.114	0.029	PRE-TERMINAL	Yajnadevam/lipi syllabic key value (not redistributed)	occurs immediately before terminal signs (modifier/inflection-like slot)	100.0% (positional class)	not estimable above null
M344	P326	W742; W743; W744	35	0.125	0.094	MEDIAL/UNSPECIFIED	Yajnadevam/lipi syllabic key value (not redistributed)	no distinctive positional behaviour (medial/unspecified)	—	not estimable above null
M178	P186	W465; W467; W468; W469; W470; W471; W472; W473; W474; W475	35	0.543	0.0	INITIAL	Yajnadevam/lipi syllabic key value (not redistributed)	text-initial sign (title/place/category-like); semantic value unknown	64.8% (positional class)	not estimable above null
M326	P112	W384; W385	35	0.25	0.071	PRE-TERMINAL	Yajnadevam/lipi syllabic key value (not redistributed)	occurs immediately before terminal signs (modifier/inflection-like slot)	90.0% (positional class)	not estimable above null
M135	P197	W482	35	0.229	0.086	PRE-TERMINAL	Yajnadevam/lipi syllabic key value (not redistributed)	occurs immediately before terminal signs (modifier/inflection-like slot)	98.0% (positional class)	not estimable above null
M186	P296	W317	33	0.469	0.094	MEDIAL/UNSPECIFIED	Yajnadevam/lipi syllabic key value (not redistributed)	no distinctive positional behaviour (medial/unspecified)	—	not estimable above null
M358	P332	W752; W760	32	0.0	0.031	PRE-TERMINAL	Yajnadevam/lipi syllabic key value (not redistributed)	occurs immediately before terminal signs (modifier/inflection-like slot)	100.0% (positional class)	not estimable above null
M103	P123; P139	W003; W013	30	0.786	0.0	NUMERAL-STROKE	Yajnadevam/lipi syllabic key value (not redistributed) | ' 3' +fish = mū-m-mīn — Parpola 1994 | numeral (stroke count) — Mahadevan 1977 sign list grouping; Parpola 1994 (e.g. p. 194)	numeral, value 3 (short strokes)	qualitative: high	not estimable above null
M233	P221	W455	29	0.143	0.107	MEDIAL/UNSPECIFIED	Yajnadevam/lipi syllabic key value (not redistributed)	no distinctive positional behaviour (medial/unspecified)	—	not estimable above null
M100	P122	W002	29	0.464	0.0	NUMERAL-STROKE	Yajnadevam/lipi syllabic key value (not	numeral, value 2 (short strokes)	qualitative: high	not estimable above null

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M	P	W	freq	init_share	final_share	structural_class	literature_values	our_function	function_confidence	phonetic_value_confidence
							redistributed) | numeral (stroke count) — Mahadevan 1977 sign list grouping; Parpola 1994 (e.g. p. 194)			
M060	P051	W226	29	0.074	0.741	TERMINAL		text-final marker (suffix/determinative/title-like); semantic value unknown	97.8% (positional class)	no proposal tested
M261	P355	W790; W850	27	0.333	0.185	MEDIAL/UNSPECIFIED	Yajnadevam/lipi syllabic key value (not redistributed) | house (place sign) — Mahadevan 2015	no distinctive positional behaviour (medial/unspecified)	—	not estimable above null
M319	P342	W165	27	0.222	0.333	MEDIAL/UNSPECIFIED	Yajnadevam/lipi syllabic key value (not redistributed)	no distinctive positional behaviour (medial/unspecified)	—	not estimable above null
M018	P023	W095	27	0.37	0.037	MEDIAL/UNSPECIFIED	Yajnadevam/lipi syllabic key value (not redistributed)	no distinctive positional behaviour (medial/unspecified)	—	not estimable above null
M153	P138		26	0.385	0.038	MEDIAL/UNSPECIFIED		no distinctive positional behaviour (medial/unspecified)	—	no proposal tested
M409	P346; P348	W170; W171	26	0.6	0.04	INITIAL	Yajnadevam/lipi syllabic key value (not redistributed)	text-initial sign (title/place/category-like); semantic value unknown	65.8% (positional class)	not estimable above null
M205			26	0.692	0.0	INITIAL		text-initial sign (title/place/category-like); semantic value unknown	92.7% (positional class)	no proposal tested
M242	P265	W621; W622	26	0.176	0.824	TERMINAL	Yajnadevam/lipi syllabic key value (not redistributed)	text-final marker (suffix/determinative/title-like); semantic value unknown	93.8% (positional class)	not estimable above null
M193	P309	W336; W337; W338	25	0.12	0.0	PRE-TERMINAL	Yajnadevam/lipi syllabic key value (not redistributed)	occurs immediately before terminal signs (modifier/inflection-like slot)	100.0% (positional class)	not estimable above null
M081	P067	W265; W267; W269; W270; W271	24	0.087	0.261	MEDIAL/UNSPECIFIED	Yajnadevam/lipi syllabic key value (not redistributed)	no distinctive positional behaviour (medial/unspecified)	—	not estimable above null
M209	P208	W500	24	0.208	0.125	BOUND-COLLOCATE	Yajnadevam/lipi syllabic key	part of a fixed collocation	71.6% (positional class)	not estimable above null

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M	P	W	freq	init_share	final_share	structural_class	literature_values	our_function	function_confidence	phonetic_value_confidence
							value (not redistributed)	(ligature/compound-like)		
M137	P240	W645	23	0.19	0.19	MEDIAL/UNSPECIFIED	Yajnadevam/lipi syllabic key value (not redistributed)	no distinctive positional behaviour (medial/unspecified)	—	not estimable above null
M096	P151	W035	22	0.143	0.381	NUMERAL-STROKE	Yajnadevam/lipi syllabic key value (not redistributed) | numeral (stroke count) — Mahadevan 1977 sign list grouping; Parpola 1994 (e.g. p. 194)	numeral, value 5 (long strokes)	qualitative: high	not estimable above null
M210	P209	W502	22	0.773	0.045	INITIAL	Yajnadevam/lipi syllabic key value (not redistributed)	text-initial sign (title/place/category-like); semantic value unknown	95.7% (positional class)	not estimable above null
M101	P140	W012	1	1.0	0.0	NUMERAL-STROKE	Yajnadevam/lipi syllabic key value (not redistributed) | numeral (stroke count) — Mahadevan 1977 sign list grouping; Parpola 1994 (e.g. p. 194)	numeral, value 2 (short strokes)	qualitative: high	not estimable above null
M105	P124	W004; W014	2	0.0	0.0	NUMERAL-STROKE	Yajnadevam/lipi syllabic key value (not redistributed) | numeral (stroke count) — Mahadevan 1977 sign list grouping; Parpola 1994 (e.g. p. 194)	numeral, value 4 (short strokes)	qualitative: high	not estimable above null
M107	P125	W005; W015	6	0.333	0.5	NUMERAL-STROKE	Yajnadevam/lipi syllabic key value (not redistributed) | numeral (stroke count) — Mahadevan 1977 sign list grouping; Parpola 1994 (e.g. p. 194)	numeral, value 5 (short strokes)	qualitative: high	not estimable above null
M108	P126	W006; W016	3	0.667	0.0	NUMERAL-STROKE	Yajnadevam/lipi syllabic key value (not redistributed) | ' 6' +fish = aru-mīn (Pleiades) — Parpola 1994 | numeral (stroke count) — Mahadevan 1977 sign list grouping; Parpola 1994 (e.g. p. 194)	numeral, value 6 (short strokes)	qualitative: high	not estimable above null
M110	P127	W007; W017	6	0.0	0.25	NUMERAL-STROKE	Yajnadevam/lipi syllabic key value (not redistributed) | ' 7' +fish = elu-mīn (Ursa Major) — Parpola 1994 | numeral (stroke count) — Mahadevan 1977	numeral, value 7 (short strokes)	qualitative: high	not estimable above null

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M	P	W	freq	init_share	final_share	structural_class	literature_values	our_function	function_confidence	phonetic_value_confidence
							sign list grouping; Parpola 1994 (e.g. p. 194)			
M114	P128	W018	7	0.429	0.143	NUMERAL-STROKE	Yajnadevam/lipi syllabic key value (not redistributed) | numeral (stroke count) — Mahadevan 1977 sign list grouping; Parpola 1994 (e.g. p. 194)	numeral, value 8 (short strokes)	qualitative: high	not estimable above null
M115	P129	W019	1			NUMERAL-STROKE	Yajnadevam/lipi syllabic key value (not redistributed) | numeral (stroke count) — Mahadevan 1977 sign list grouping; Parpola 1994 (e.g. p. 194)	numeral, value 9 (short strokes)	qualitative: high	not estimable above null
M116	P129	W019	2	0.5	0.0	NUMERAL-STROKE	Yajnadevam/lipi syllabic key value (not redistributed) | numeral (stroke count) — Mahadevan 1977 sign list grouping; Parpola 1994 (e.g. p. 194)	numeral, value 9 (short strokes)	qualitative: high	not estimable above null
M118	P130	W020	1			NUMERAL-STROKE	Yajnadevam/lipi syllabic key value (not redistributed) | numeral (stroke count) — Mahadevan 1977 sign list grouping; Parpola 1994 (e.g. p. 194)	numeral, value 10 (short strokes)	qualitative: high	not estimable above null
M111	P136	W048; W049	4	0.5	0.0	NUMERAL-STROKE	Yajnadevam/lipi syllabic key value (not redistributed)	stroke variant (bracketed strokes); numeral status uncertain	qualitative: low–moderate	not estimable above null
M113	P136	W048; W049	17	0.0	0.118	NUMERAL-STROKE	Yajnadevam/lipi syllabic key value (not redistributed)	stroke variant (bracketed strokes); numeral status uncertain	qualitative: low–moderate	not estimable above null
M117	P137	W051	1	1.0	0.0	NUMERAL-STROKE	Yajnadevam/lipi syllabic key value (not redistributed)	stroke variant (roofed strokes); numeral status uncertain	qualitative: low–moderate	not estimable above null
M119	P141	W025; W026; W027; W028; W029	13	1.0	0.0	NUMERAL-STROKE	Yajnadevam/lipi syllabic key value (not redistributed)	stroke variant (slanted strokes); numeral status uncertain	qualitative: low–moderate	not estimable above null
M120	P141	W025; W026; W027; W028; W029	2	1.0	0.0	NUMERAL-STROKE	Yajnadevam/lipi syllabic key value (not redistributed)	stroke variant (slanted strokes); numeral status uncertain	qualitative: low–moderate	not estimable above null
M122	P143	W057	1	1.0	0.0	NUMERAL-STROKE	Yajnadevam/lipi syllabic key	stroke variant (bracketed strokes);	qualitative: low–moderate	not estimable above null

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M	P	W	freq	init_share	final_share	structural_class	literature_values	our_function	function_confidence	phonetic_value_confidence
							value (not redistributed)	numeral status uncertain		
M088	P146		1			NUMERAL-STROKE		stroke variant (long+short strokes); numeral status uncertain	qualitative: low–moderate	no proposal tested

Appendix B. Hand-sourced sign-value proposals and held-out outcomes

Source. Rows are taken from `outputs/claims/all_proposals.csv`, which also holds the 714 Yajnadevam and 99 Field-Cady key rows; their per-sign results are in `outputs/claims/loo_phonetic_per_sign.csv`.

Pages. Page numbers are book pages. For Parpola (1994), the scan offsets were verified individually; Mahadevan (2015) is unpaginated.

Mapping column. This gives the confidence that the sign named by the proponent is the M77 sign listed.

Held-out test column. This tests the role implied by the claim on 641 texts from sites other than Mohenjo-daro and Harappa. It does **not** test the sound value.

#	Sign (M77)	Value	Meaning	Proponent	Source: page	Mapping	Held-out test (other sites)
1	M059	mīn	‘fish’ = homophone ‘star’	Parpola	Parpola 1994: 181; 187-188	high	no distributional prediction tested
2	M108/M109+M059	aṟu-mīn	‘six stars’ = Pleiades	Parpola (after Heras)	Parpola 1994: 194	high	consistent: adjacent pair obs 3 vs 0.63, q=0.013
3	M102/M103+M059	mū-m-mīn	‘three stars’	Parpola	Parpola 1994: 194	high	consistent: adjacent pair obs 3 vs 0.61, q=0.025
4	M110/M112+M059	eḻu-mīn	‘seven stars’ = Ursa Major	Parpola	Parpola 1994: 194	high	untestable held-out (no text has both); in-sample 1 vs 1.39, p=.82
5	M065	mēy-mīn	‘roofed fish’ = ‘black star’ = Saturn	Parpola	Parpola 1994: 197	medium	no distributional prediction tested
6	M403	muruku	ring/bangle = ‘Muruku’ (god)	Parpola	Parpola 1994: 276 (Fig. 15.2 no. 11)	low	not testable (sign unmapped or class-level)
7	M103?+M403	cūl + muruku	hearth of three stones + bangle	Parpola	Parpola 1994: 276 (Fig. 15.2 no. 12)	low	not testable (sign unmapped or class-level)
8	M403+M087	Muruku-Vēḷ	bangle + intermediate space	Parpola	Parpola 1994: 276 (Fig. 15.2 no. 14)	low	consistent: adjacent pair obs 4 vs 0.98, q=0.012

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#	Sign (M77)	Value	Meaning	Proponent	Source: page	Mapping	Held-out test (other sites)
9	M087	vēḷ(i)	‘intermediate space’ (two long strokes) / ‘white’	Parpola	Parpola 1994: 276-277 (Fig. 15.2 nos. 14-15)	medium	no distributional prediction tested
10	M087+M059	vēḷ-mīn	‘white star’ = Venus	Parpola	Parpola 1994: 277 (Fig. 15.2 no. 15)	medium	consistent: adjacent pair obs 7 vs 1.75, q=0.0013
11	fig-tree sign (unmapped)+M059	vaṭa-mīn	‘banyan star’ = North Star	Parpola	Parpola 1994: 277 (Fig. 15.2 no. 16)	none	not testable (sign unmapped or class-level)
12	fig-tree sign+M087	vaṭa + vēḷ(i)		Parpola	Parpola 1994: 277 (Fig. 15.2 no. 17)	none	not testable (sign unmapped or class-level)
13	M104+fig-tree sign	nāl + vaṭa	‘hanging rope’	Parpola	Parpola 1994: 277 (Fig. 15.2 no. 18)	none	not testable (sign unmapped or class-level)
14	crab sign (unmapped)	kōḷ	‘grasper’ = ‘planet’	Parpola	Parpola 1994: 277 (Fig. 15.2 nos. 19-21)	none	not testable (sign unmapped or class-level)
15	M328	(no phonetic value given)	pot U	Parpola	Parpola 1994: 277 (Fig. 15.2 no. 22)	medium	no distributional prediction tested
16	M001	āḷ / āṇ	‘man; servant’	Parpola	Parpola 1994: 188; 277 (Fig. 15.2 no. 23)	high	no distributional prediction tested
17	M342	ā (+ -tu)	‘(head of) cow’ = genitive/possessive suffix *-ā	Parpola	Parpola 1994: 188; 277 (Fig. 15.2 no. 24)	high	consistent: text-final obs 131 vs 45.42, q=0.0013
18	fish-with-dot (unmapped)	pottu	dot = ‘pottu’	Parpola	Parpola 1994: 264	none	not testable (sign unmapped or class-level)
19	M342	-at(tu)	Proto-Dravidian oblique-case suffix, via rebus ati/atti fig	Knorozov / Soviet team	via Parpola 1994: 59-60	high	consistent: text-final obs 131 vs 45.42, q=0.0013
20	star sign (unmapped)	sukka	‘star’	Knorozov / Soviet team	via Parpola 1994: 59-60	none	not testable (sign unmapped or class-level)
21	M342	(priest)	jar = ‘priest’	Mahadevan (early)	via Parpola 1994: 60	high	no distributional prediction tested
22	spear sign (unmapped; M211?)	(soldier)	‘soldier’	Mahadevan (early)	via Parpola 1994: 60	low	not testable (sign unmapped or class-level)
23	arrow + half fig leaf (unmapped)	amp-ara	‘public space’	Fairservis	via Parpola 1994: 60	none	not testable (sign

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#	Sign (M77)	Value	Meaning	Proponent	Source: page	Mapping	Held-out test (other sites)
							unmapped or class-level)
24	M108/M109+M059	(Pleiades)	‘6’+fish = Pleiades (first proposed)	Heras	via Parpola 1994: 59; 277	high	consistent: adjacent pair obs 3 vs 0.63, q=0.013
25	M342	-(a)nṟu	masculine singular suffix	Mahadevan	Mahadevan 2015: unpag. (Paradigm of gender)	high	consistent: text-final obs 131 vs 45.42, q=0.0013; paradigm with M211 not supported (p=.32)
26	M211	ampu > -(a)mpu	arrow ‘ampu’ = non-masculine singular suffix (after Mahadevan 1998)	Mahadevan	Mahadevan 2015: unpag. (Interpreting the arrow sign)	high	consistent: text-final obs 38 vs 12.46, q=0.0013; paradigm with M342 not supported (p=.32)
27	four-stroke modifier (on M060)	-(a)r	epicene plural	Mahadevan	Mahadevan 2015: unpag.	medium	no distributional prediction tested
28	M267+M099	aka-tt-(i)	‘lord of the house’	Mahadevan	Mahadevan 2015: unpag.	high	consistent: adjacent pair obs 56 vs 15.92, q=0.0013
29	M267	aka-(tt-i)	‘palace (fort, court)’	Mahadevan	Mahadevan 2015: unpag. (place-sign chart)	high	consistent: text-initial obs 66 vs 21.62, q=0.0013
30	M284	pāḷ(-i)	‘city’	Mahadevan	Mahadevan 2015: unpag. (place-sign chart)	high	not supported: text-initial obs 0 vs 0.58, q=1
31	M264	ampal(a)	‘assembly (place at four-road junction)’	Mahadevan	Mahadevan 2015: unpag. (place-sign chart)	medium	not supported: text-initial obs 1 vs 0.15, q=0.23
32	M150	kēr(-i)	‘exclusive quarters’	Mahadevan	Mahadevan 2015: unpag. (place-sign chart)	medium	consistent: text-initial obs 4 vs 0.9, q=0.0079
33	M149	pāṭ(-i)	‘ordinary quarters’	Mahadevan	Mahadevan 2015: unpag. (place-sign chart)	medium	not supported: text-initial obs 2 vs 1.02, q=0.37
34	M267 variant (not separately cropped)	cūḷ(-i)	palace row, second value (‘counsel’)	Mahadevan	Mahadevan 2015: unpag. (place-sign chart)	low	not testable (sign unmapped or class-level)
35	M261/M373	(house)	‘house’	Mahadevan	Mahadevan 2015: unpag.	medium	not supported: text-initial obs 3 vs 3.84, q=0.96

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#	Sign (M77)	Value	Meaning	Proponent	Source: page	Mapping	Held-out test (other sites)
36	M047/M048	muruku	seated deity = Muruku	Mahadevan	Mahadevan 2015: unpag. (Fig. 4C labels “47”, “48”)	high	no distributional prediction tested
37	M012	poṛ-ay	‘bearer (sustainer)’	Mahadevan	Mahadevan 2015: unpag. (Indus heritage chart)	medium	no distributional prediction tested
38	M015	cāt(t)-poṛ-ay	‘jar-bearer (food-bearer)’	Mahadevan	Mahadevan 2015: unpag. (Indus heritage chart)	high	no distributional prediction tested
39	M014	ey-poṛ-ay	‘arrow-bearer (arms-bearer)’	Mahadevan	Mahadevan 2015: unpag. (Indus heritage chart)	high	no distributional prediction tested
40	M267+M342 (W861+W740)	‘cow at the city’	sequence reading of H-176	Wells	Wells 2015: 93 (Fig. 6.12 caption, TOC only)	high	not supported as a unit: adjacent pair obs 3 vs 5.88, q=1
41	phrase-final signs (class)	(functional morphemes)	PF signs are functional/terminal markers	Mukhopadhyay	Mukhopadhyay 2019: whole article	n/a	not testable (sign unmapped or class-level)
42	numeral/stroke signs (class)	(quantifiers)	stroke signs quantify	Mukhopadhyay	Mukhopadhyay 2019: whole article	n/a	not testable (sign unmapped or class-level)

Appendix C. Full sign map

Files. The complete map has 1,132 rows: 418 M and 714 W.

- `outputs/signmap/sign_map_full_public.csv`; the SQLite table `sign_map_full`; summary in `sign_map_full_summary.json`
- concordance scoring: `outputs/signmap/concordance_links_scored.csv`
- the AI-made visual check (not a human check): `handcheck_pairs.csv`
- look-alikes: `outputs/comparanda/lookalikes_top3.csv`

Columns

Group	Columns
Identity	scheme, sign, descriptions
Parpola IDs	P, parpola_variants
Allographs	allographs sharing a P
Concordance	W_links or M_links with link confidence; W_hungarian; W_proposed_if_unlinked
Distribution	freq; solo, initial, medial and final shares; top left/right neighbours; sites; object types
Classes	provisional_class (binomial p); structural_class and class_confidence (session 3)
Glyphs	glyph file (private)
Look-alikes	lookalike_
Claims	published_claims; claim_tests_heldout

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Group	Columns
Segmentation	BE_units
Phonetic keys	Yajnadevam and Field-Cady values with leave-one-out p
Yajnadevam decomposition	lipi_canonical_decomposition (W rows, labelled as Yajnadevam’s)

The table is too wide to print here.

PREPRINT